

A MAJOR PROJECT REPORT

on

FIRE FIGHTING ROBOT USING ARDUINO WITH ALERT

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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**DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING)**

**VIGNAN'S INSTITUTE OF MANAGEMENT AND TECHNOLOGY
FOR WOMEN**

Accredited to NAAC A⁺ and NBA (CSE & ECE)

(Affiliated to Jawaharlal Nehru Technological University Hyderabad) Kondapur
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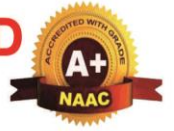


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CERTIFICATE

This is to certify that project work entitled “**FIRE FIGHTING ROBOT USING ARDUINO WITH ALERT**” submitted by **D.Likhitha Mahalaxmi (20UP1A6611), G.Shravani (20UP1A6614), M.Shruthi (20UP1A6632), M.Sreeja (21UP5A6602)** in the partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in CSE (Artificial Intelligence and Machine Learning)** VIGNAN'S INSTITUTE OF MANAGEMENT AND TECHNOLOGY FOR WOMEN is a record of bonafide work carried by them under my guidance and supervision. The results embodied in this project report have not been submitted to any other University or institute for the award of any degree.

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DECLARATION

We hereby declare that project entitled “**FIRE FIGHTING ROBOT USING ARDUINO WITH ALERT**” is bonafied work duly completed by us. It does not contain any part of the project or that is submitted by any other candidate to this or another institute of the university. All such materials that have been obtained from other sources have been duly acknowledged.

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ABSTRACT

The main goal of this project is to develop a robotic vehicle which is used to find and fight fire remotely in an event of any major fire hazard particularly in industries. Major fire accidents do occur in industries like nuclear power plants, petroleum refineries, gas tanks, chemical factories and other large-scale fire industries resulting in quite serious consequences. Thousands of people have lost their lives in such mishaps. Therefore, this project is enhanced to control fire through a robotic vehicle with the advancement in the field of Robotics, human intervention is becoming less every day and robots are used widely for purpose of safety. In our day-to-day life fire accidents are very common and sometime it becomes very difficult for fireman to save human life. In such case firefighting robot comes in picture. The fire extinguishing robotic vehicle can be controlled wirelessly Bluetooth communication. The proposed vehicle has a water jet spray which is capable of sprinkling water. The sprinkler can be moved towards the required direction.

This project describes a new economical solution of robot control systems. The controlling devices of the whole system are Microcontrollers, wireless transceiver modules, water jet spray, DC motors flame sensor and buzzer are interfaced to Microcontroller. When the user fed the commands through a remote-controlled device, the microcontroller interfaced to it reads the command and sends relevant data of that command wirelessly using transceiver module. This data is received by the transceiver module on the robot and fed sit to microcontroller which acts accordingly on motors and pump. The complete system consists of two subsystems transmitter section and the receiver section. This project controls left, right, forward and backward movement of robot wirelessly. At the receiver side of robot microcontroller is also used.

TABLE OF CONTENTS

CONTENTS	PAGE NO
1. INTRODUCTION	1-2
2. LITERATURE SURVEY	3-4
3. EXISTING SYSTEM	5-6
4. PROPOSED SYSTEM	7-9
5. SYSTEM ARCHITECTURE	10-41
5.1 Block diagram Overview	10
5.2 Power Supply	10
5.3 Components	16
6. ARDUINO CONTROLLER	42-49
7. IMPLEMENTATION	50-69
7.1 Software requirements	50
7.2 The Interface	50
7.3 Design Considerations	51
7.4 Code Implementation	52
7.4 Arduino Compiling	65
8. METHODOLOGY WITH WORKING	70-72
9. ADVANTAGES AND LIMITATIONS	73
10. CONCLUSION	74-75
11. BIBILOGRAPHY	76-77

LIST OF FIGURES

Name of Figures	Page No
5.1 Block Diagram of fire fighting robot	10
5.2.1 Basic block diagram of fixed regulated power supply	11
5.2.2 Transformer and Center tapped Transformer	11
5.2.3 Half wave Rectifier	12
5.2.4 Full Wave Rectifier	13
5.2.5 Bridge Rectifier	15
5.2.6 Bridge Rectifier with positive and negative half cycles	15
5.3.1 Filter Capacitor	16
5.3.2 Voltage Regulator	18
5.3.3 Circuit Diagram of power supply	18
5.3.4 LCD Pin diagram	20
5.3.5 Circuit Diagram of LCD	22
5.3.6 Bluetooth Module	26
5.3.7 HC 05 Module	28
5.3.8 Diagram of Gas sensor	30
5.3.9 Relay	33
5.3.10 DC Motor	33
5.3.11 Simple electrical diagram of DC Motor	34
5.3.12 Operation of a DC Motor	35
5.3.13 Diagram of DC shunt motor	36
5.3.14 Diagram of DC series motor	37
5.3.15 Diagram of DC series motor graph representation	37
5.3.16 Diagram of DC compound motor	38
5.3.17 L293D IC	39
5.3.18 L293D motor driver pin diagram	40
5.3.19 Internal structure of L293D	40
6.1 Hardware Image	43

6.2 Back side of Module	45
6.3 Arduino board	46
6.4 Pin explanation	49
7.2.1 Tool bar necessary for the interface	50
8.1 Hardware kit image	70
8.2 Smoke sensor detected time	71
8.3 Fire sensor detected indication	71

LIST OF TABLES

Name of Tables	Page No
5.3.1 Specifications of IC785	19
5.3.2 Character LCD pins with Microcontroller	20
5.3.3 Pin Description of Bluetooth Module	27
5.3.4 Pin Description of Flame sensor Module	29

CHAPTER 1

INTRODUCTION

Fire incidents can pose significant threats to lives and properties, emphasizing the need for efficient firefighting mechanisms. With advancements in technology, the integration of robotics into firefighting operations has become increasingly prevalent. In this context, the development of a fire-fighting robot utilizing Arduino technology, coupled with alert functionalities, emerges as a promising solution to enhance response capabilities. The Arduino platform serves as the foundation for the fire-fighting robot, providing a versatile and customizable framework for implementation. Equipped with sensors and actuators, the robot is designed to detect and combat fire hazards autonomously. Key components include temperature sensors for fire detection, motors for locomotion, and a water spraying mechanism for extinguishing flames.

The operation of the fire-fighting robot revolves around its ability to sense and respond to fire incidents effectively. Temperature sensors are strategically positioned to monitor ambient temperatures continuously. Upon detecting a significant rise in temperature indicative of a fire outbreak, the robot initiates its response protocol. In response to a detected fire, the robot employs its locomotion system to navigate towards the source of the heat. Utilizing predefined algorithms, it calculates the optimal path to reach the fire-affected area swiftly. This autonomous navigation capability ensures timely intervention, minimizing the spread of fire and reducing potential damages.

Upon reaching the vicinity of the fire, the robot activates its water spraying mechanism to extinguish the flames. The water reservoir, integrated with the robot, supplies the necessary firefighting agent to combat the fire effectively. Controlled by Arduino-based algorithms, the spraying mechanism delivers a targeted stream of water to the affected area, aiding in suppression efforts. In addition to its primary firefighting functions, the robot incorporates alert features to notify relevant authorities and stakeholders about the ongoing fire incident. Upon detecting a fire, the robot triggers an alert system that initiates automated messages to predefined contacts, including emergency services, building occupants.

The integration of alert functionalities enhances the responsiveness of firefighting operations by ensuring prompt communication and coordination. Emergency responders receive real-time notifications, allowing them to initiate appropriate measures and mobilize resources effectively.

Simultaneously, building occupants are promptly informed about the situation, facilitating orderly evacuation procedures and enhancing overall safety. The implementation of alert capabilities leverages GSM (Global System for Mobile Communications)

modules interfaced with Arduino. These modules enable the robot to establish cellular communication networks, facilitating the transmission of alert messages and calls. Predefined contact information is stored within the robot's memory, enabling seamless activation of the alert system during fire emergencies.

Objectives of Fire Fighting Robot

- Develop an autonomous fire-fighting robot for indoor environments.
- Implement a fire suppression mechanism for efficient extinguishing.
- Enhance fire safety and response efficiency through automation.
- Utilize Arduino Uno for control and programming flexibility.
- Allow customization and future expansion.

CHAPTER 2

LITERATURE SURVEY

For many years robotics have become popular in many ways because of its variety of design and technological advancement. Our motivation to design a fire fighter robot is to help the community in effectively firefighting by sensing the fire and quickly acting without heavy losses of property or life. In some projects a tracking line is assigned for the robot to follow these paths to extinguish the fire. In some designs ultrasonic sensors are used. When it comes to simulation of this design it will make it difficult to implement it in real time situations. From the different projects reviewed, certain objectives helped us to choose a popular technique to fight fire efficiently by controlling it remotely by wireless technology application and using two main sensors on Temperature sensor, GAS sensor due to their ability to detect fire and fire extinguisher pump.

This is mainly because they can be used in situations that are too dangerous for any individual to involve themselves in. In our project, we develop a robot that is able to locate and extinguish fire in a given environment. The robot navigates the area and avoids any obstacles it faces in its excursion. Hear about robots, we think of science fiction novels and sci-fi movies. It is due to the fact that we do not know how to create robots of high intelligence.

1 Systems, Man and Cybernetics, 2006. SMC '06. IEEE International Conference on 8-11 Oct.2006 The adaptive fusion method is proposed for fire detection of firefighting robot.

2 “Fire fighting robot” by S. Kannan, R. Karthikeyan, and S. Sathish Kumar, Chennai, MAY 2006. Design a Fire fighting robot using embedded system.

3 “Fire Fighting Robot” invented by Hadi A. Al-Azemi, Feb 26, 2013. This invention relates to a firefighting robot with a caterpillar drive assembly including a left and right drive unit and an engine for each drive unit.

4 “Fire Fighting Robot” Choong, Cheat Sun 2015-11. This project mainly to portray basic function of a real big scale firefighting robot into a small-scale robot.

Study	Description	Key Features
[1] Systems, Man and Cybernetics, 2006. SMC '06. IE IEEE International Conference on	The adaptive fusion method is proposed for fire detection of firefighting robot.	- Adaptive fusion method for fire detection
[2] "Fire fighting robot" by S. Kannan, R. Karthikeyan, and S. Sathish Kumar, Chennai, MAY 2006	Design of a Fire fighting robot using embedded system.	- Embedded system based design
[3] "Fire Fighting Robot" invented by Hadi A. Al-Azemi, Feb 26, 2013	This invention relates to a firefighting robot with a caterpillar drive assembly including a left and right drive unit and an engine for each drive unit.	- Caterpillar drive assembly with left and right drive units
[4] "Fire Fighting Robot" Choong, Cheat Sun 2015-11	This project mainly to portray basic function of a real big scale firefighting robot into a small-scale robot.	- Small-scale robot with basic firefighting functions

CHAPTER 3

EXISTING SYSTEM

Fire accidents pose significant threats to lives and property, underscoring the critical importance of prompt and effective fire management measures. As such, the integration of advanced technologies, particularly fire-fighting robots, has emerged as a crucial strategy in mitigating these risks. Leveraging Arduino, an open-source electronics platform renowned for its versatility and flexibility, alongside alert functionalities, presents a robust solution for enhancing fire management capabilities.

The fire-fighting robot system comprises a suite of components designed to ensure efficient operation in the face of fire emergencies. At its heart lies Arduino, serving as the central control unit responsible for orchestrating the robot's actions and managing communication protocols for alert dissemination. Equipped with an array of sensors, including temperature and smoke detectors, the robot is empowered to swiftly detect fire outbreaks within its vicinity, initiating timely responses to mitigate potential damages.

Upon detecting a fire incident, the robot swiftly activates its response protocol, which typically involves deploying its extinguishing mechanisms. These mechanisms may vary depending on the nature of the fire, ranging from water spraying to foam dispensing, each tailored to combat specific fire types effectively. Arduino's inherent flexibility allows for seamless integration with these mechanisms, enabling precise control and coordination of firefighting efforts.

In addition to its autonomous firefighting capabilities, the system incorporates alert functionalities to notify designated authorities or individuals about the emergency situation. Arduino facilitates the integration of GSM modules, enabling the robot to establish cellular communication networks for the transmission of alerts. When triggered by a fire incident, the Arduino activates the GSM module, initiating predefined alert messages. These alerts can be directed to emergency services, building occupants, or facility managers, ensuring swift response and coordinated action.

Moreover, the system can include location information in the alerts, providing responders with crucial details to pinpoint the exact location of the fire. By incorporating modules or location

tracking algorithms, the system enhances its effectiveness in guiding responders to the scene of the emergency, facilitating timely intervention and containment efforts.

The integration of alerts significantly enhances the system's effectiveness by facilitating timely communication and mobilization of resources. Authorities can promptly assess the situation based on the received alerts and initiate appropriate responses, such as dispatching fire brigades or implementing evacuation protocols. Furthermore, the system can incorporate feedback mechanisms to provide real-time updates on the firefighting progress.

Sensors onboard the robot continuously monitor parameters such as temperature, air quality, and humidity, relaying this information back to the control unit via Arduino. Based on these insights, the system can dynamically adjust its firefighting strategy for optimal efficiency, ensuring the effective containment and extinguishment of the fire while minimizing risks to personnel and property.

The integration of Arduino-based fire-fighting robots with alert functionalities represents a robust and versatile solution for enhancing fire management capabilities. By leveraging advanced technologies and communication protocols, the system enables swift detection, response, and containment of fire emergencies, ultimately safeguarding lives and property against devastating consequences.

CHAPTER 4

PROPOSED SYSTEM

The basic theme of this project is to sense the environmental fire and extinguish it with the help of a water pump. The Arduino UNO Microcontroller board based on the ATmega328P. The ATmega328P is good platform for robotics application. Thus the real time fire extinguishing can be performed. The Arduino software runs on different platforms such as mac, windows and Linux. Simple and clear programming is possible in case of Arduino software.

The Arduino libraries play a major role in making the programming easier by providing wider range of libraries. There are many built in libraries available in the Arduino software and it allows to add additional libraries that are available in the open source for download. Adding of new boards to Arduino software is possible. Since, Arduino C is derived from C and C++ programming and is much easier when compared other controller programming. The microcontroller in turn controls the extinguishing system. The Operating Voltage of the controller is 5V and the Clock Speed is 16 MHz, and the recommended Input Voltage 7-12V, whereas the limitation of Input Voltage between 6-20V. The Dual-tone multi- frequency signaling (DTMF) is an in-band telecommunication signaling system which uses the voice frequency band over telephone lines between telephone equipment and other communications devices and switching centers.

Here, the IC MT8870DE, a touch tone decoder IC is used. The main aim of this project is to develop a DTMF controlled fire extinguishing robot which detects the fire location and extinguish fire by using sprinklers on triggering the pump. The directions of movement of the robot are described by the motor driver board. It is used to give high voltage and high current is given as an output to run the motors which are used in the project for the movement of the robot.

In this project a simple DC motor is used for the rotation of the wheel which are responsible for the movement of the robot. DC motors usually convert electrical energy into mechanical energy. To extinguish the fire a pump is used to pump the water on to the flame. A simple motor is used to pump the water. The pumping motor in extinguishing system controls the flow of water coming out of pumping.

COMPONENTS USED

1. MQ-2 Gas Sensor:

- Detects various gases in the air such as LPG, propane, methane, hydrogen, alcohol, smoke, and carbon monoxide.
- Operates based on changes in resistance when exposed to different gas concentrations.
- Widely used in applications such as gas leakage detection, industrial safety, and air quality monitoring.

2. Flame Sensor:

- Specifically designed to detect the presence of flames, including smokeless liquid and smoke that can ignite.
- Utilizes sensors capable of detecting specific wavelengths of light emitted by flames.
- Commonly used in fire alarm systems, industrial furnaces, and gas appliances for rapid fire detection and prevention.

3. SIM800L:

- A miniature cellular module enabling GSM/GPRS communication.
- Allows for sending and receiving messages, making and receiving voice calls, and internet connectivity through GPRS.
- Suitable for applications requiring remote communication capabilities, such as IoT devices, security systems, and remote monitoring systems.

4. Servo Motor:

- Converts electrical signals from a controller into precise rotational or angular displacement.
- Provides accurate control over the position or velocity of the motor's output shaft.
- Commonly used in robotics, automation, and remote-controlled devices for precise motion control.

5. Relay Module:

- Switches electrical devices on and off based on control signals.
- Isolates control circuits from higher voltage or current devices, ensuring safety for low-power control components.
- Used in various applications such as home automation, industrial control systems, and automotive electronics for controlling high-power devices with low-power control signals.

6. Motor Driver:

- Controls the motion and direction of a motor by regulating the flow of current.
- Typically used with DC motors or stepper motors to control speed and direction.
- Provides protection features such as overcurrent protection and thermal shutdown, ensuring the safety and reliability of motor operations.

Advantages of Fire Fighting Robot

1. The robot will be used at places where it is dangerous for humans to enter.
2. Capability of sensing accurately with increased flexibility.
3. Reduce human effort.
4. Reliable and economical.
5. It reduces the time delay in reaching fire affected area.
6. It reduce the errors and the limitations that are faced by human fire fighters.
7. Sensors have long life time and less cost.

CHAPTER 5

SYSTEM ARCHITECTURE

5.1 BLOCK DIAGRAM OVERVIEW

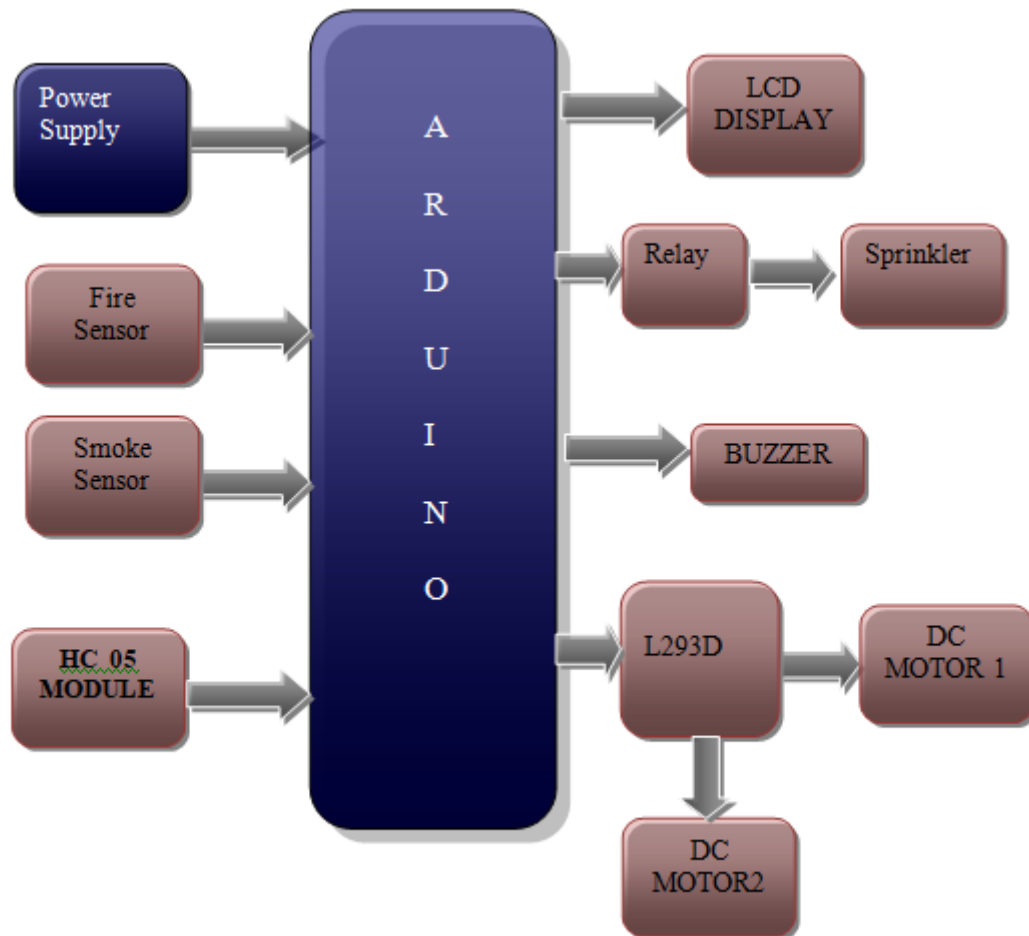


Fig 5.1 Block diagram of fire fighting robot

5.2 POWER SUPPLY

All digital circuits require regulated power supply. In this article we are going to learn how to get a regulated positive supply from the mains supply.

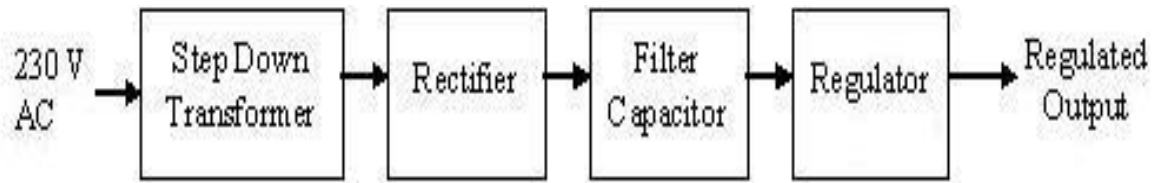


Fig 5.2.1 basic block diagram of a fixed regulated power supply

TRANSFORMER

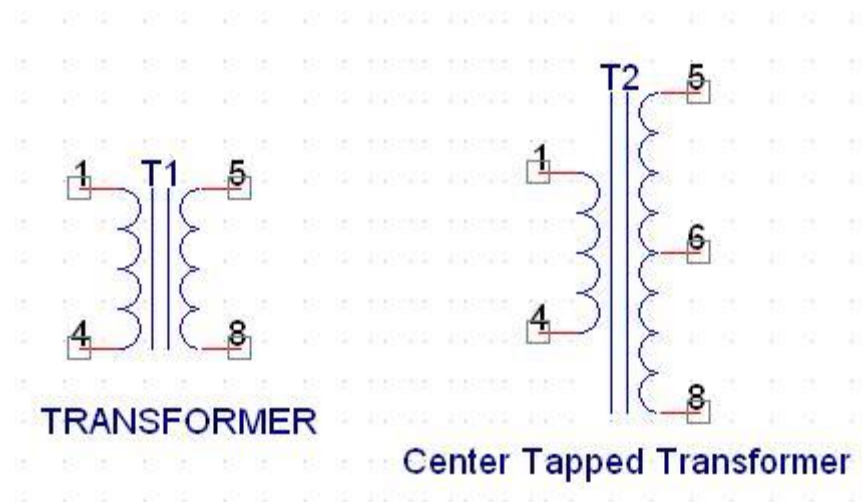


Fig 5.2.2 Transformer and Center Tapped Transformer

A transformer consists of two coils also called as “WINDINGS” namely PRIMARY & SECONDARY.

They are linked together through inductively coupled electrical conductors also called as CORE. A changing current in the primary causes a change in the Magnetic Field in the core & this in turn induces an alternating voltage in the secondary coil. If load is applied to the secondary then an alternating current will flow through the load. If we consider an ideal condition then all the energy from the primary circuit will be transferred to the secondary circuit through the magnetic field.

$$P_{\text{primary}} = P_{\text{secondary}}$$

So

$$I_p V_p = I_s V_s$$

The secondary voltage of the transformer depends on the number of turns in the Primary as well as in the secondary.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Rectifier

A rectifier is a device that converts an AC signal into DC signal. For rectification purpose we use a diode, a diode is a device that allows current to pass only in one direction i.e. when the anode of the diode is positive with respect to the cathode also called as forward biased condition & blocks current in the reversed biased condition.

Rectifier can be classified as follows:

1. Half Wave rectifier.
2. Full wave rectifier.
3. Bridge rectifier

1) Half Wave rectifier

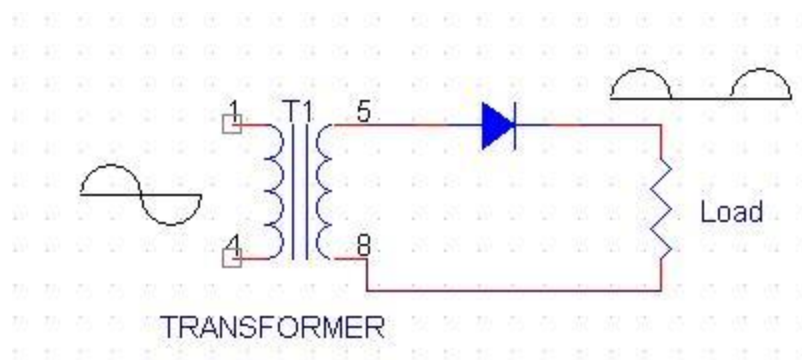


Fig 5.2.3 Half Wave rectifier

A half-wave rectifier for a firefighting robot would likely be integrated into its power supply system to ensure stable DC power for its electronics. The rectifier's primary function is to

convert alternating current (AC) from a power source, such as a battery or generator, into direct current (DC). In the context of the firefighting robot, the half-wave rectifier would consist of a diode connected in series with a load (the robot's electronics). During the positive half-cycle of the AC input voltage, the diode conducts and allows current to flow through the load, providing power. However, during the negative half-cycle, the diode blocks current flow, resulting in a pulsating DC output. While a half-wave rectifier is simple and inexpensive, it suffers from low efficiency and produces a significant amount of ripple in the output voltage. Despite its limitations, it can still serve adequately for low-power applications like powering sensors or control circuits in the firefighting robot, provided the ripple voltage is acceptable for the system's operation.

This is the simplest type of rectifier as you can see in the diagram a half wave rectifier consists of only one diode. When an AC signal is applied to it during the positive half cycle the diode is forward biased & current flows through it. But during the negative half cycle diode is reverse biased & no current flows through it. Since only one half of the input reaches the output, it is very inefficient to be used in power supplies.

2) Full wave rectifier

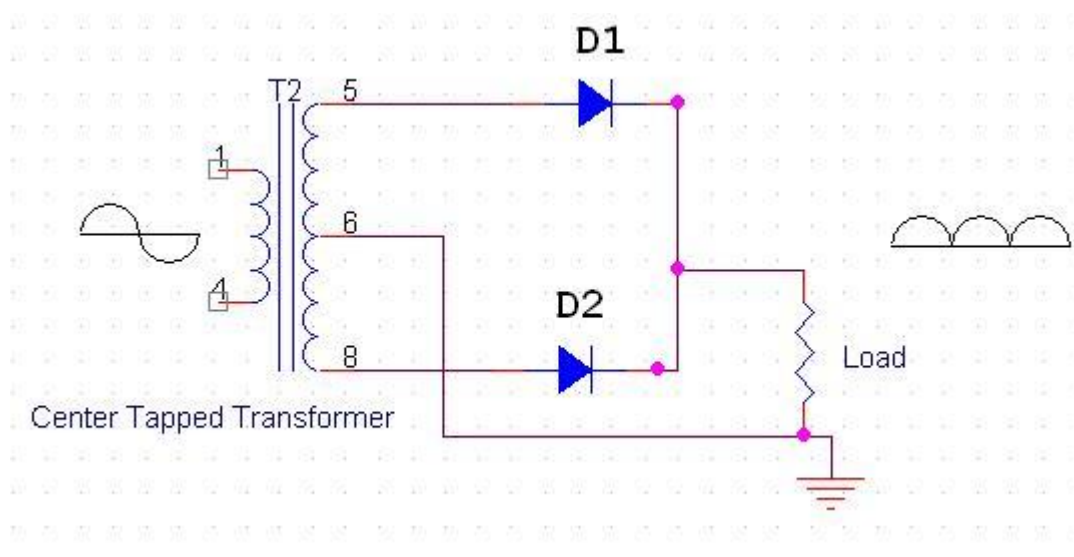


Fig 5.2.4 Full wave rectifier

A full-wave rectifier for a firefighting robot plays a crucial role in converting alternating current (AC) from its power source into direct current (DC) to power its electronics reliably. Unlike a half-wave rectifier, which only utilizes one half of the AC input cycle, a full-wave rectifier utilizes both halves, resulting in a smoother DC output with reduced ripple. In the context of the firefighting robot, the full-wave rectifier typically consists of a bridge rectifier configuration, composed of four diodes arranged in a bridge arrangement. This configuration ensures that current flows in the same direction through the load (the robot's electronics) during both halves of the AC input cycle, effectively doubling the frequency of the output waveform compared to a half-wave rectifier.

The advantages of a full-wave rectifier include higher efficiency and a more stable output voltage compared to a half-wave rectifier, making it suitable for powering sensitive electronics and actuators in the firefighting robot. By providing a smoother DC output, the full-wave rectifier contributes to the robot's overall performance and reliability in challenging firefighting scenarios, where consistent power delivery is essential for mission success. Half wave rectifier is quite simple but it is very inefficient, for greater efficiency we would like to use both the half cycles of the AC signal. This can be achieved by using a center tapped transformer i.e. we would have to double the size of secondary winding & provide connection to the center. So during the positive half cycle diode D1 conducts & D2 is in reverse biased condition. During the negative half cycle diode D2 conducts & D1 is reverse biased. Thus we get both the half cycles across the load.

One of the disadvantages of Full Wave Rectifier design is the necessity of using a center tapped transformer, thus increasing the size & cost of the circuit. This can be avoided by using the Full Wave Bridge Rectifier.

3) Bridge rectifier

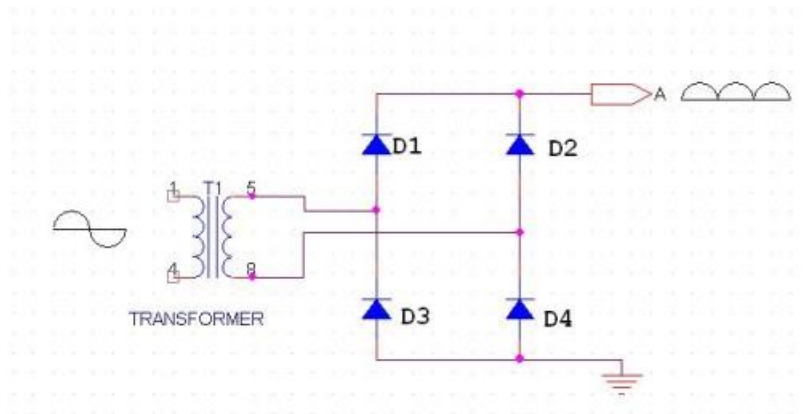


Fig 5.2.5 Bridge rectifier

As the name suggests it converts the full wave i.e. both the positive & the negative half cycle into DC thus it is much more efficient than Half Wave Rectifier & that too without using a center tapped transformer thus much more cost effective than Full Wave Rectifier. Full Bridge Wave Rectifier consists of four diodes namely D1, D2, D3 and D4. During the positive half cycle diodes D1 & D4 conduct whereas in the negative half cycle diodes D2 & D3 conduct thus the diodes keep switching the transformer connections so we get positive half cycles in the output.

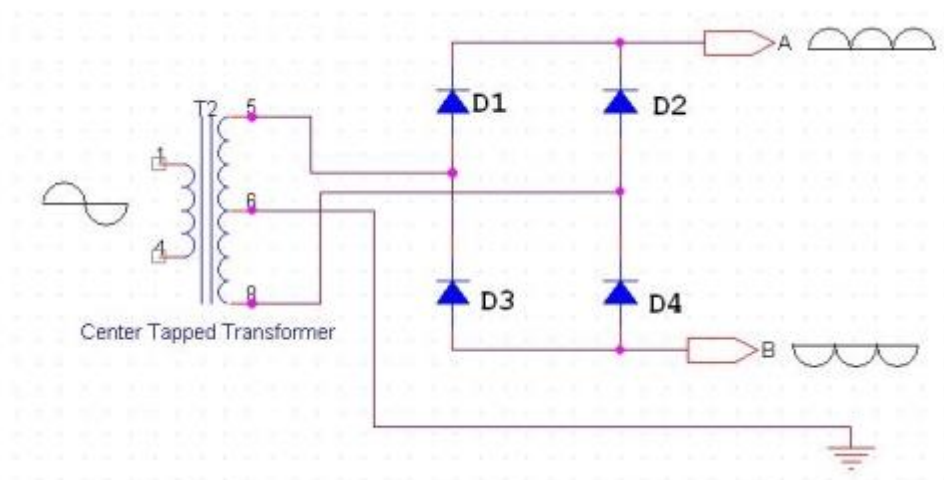


Fig 5.2.6 Bridge rectifier with positive and negative half cycles

If we use a center tapped transformer for a bridge rectifier we can get both positive & negative half cycles which can thus be used for generating fixed positive & fixed negative voltages.

5.3 COMPONENTS

FILTER CAPACITOR

Even though half wave & full wave rectifier give DC output, none of them provides a constant output voltage. For this we require to smoothen the waveform received from the rectifier. This can be done by using a capacitor at the output of the rectifier this capacitor is also called as “FILTER CAPACITOR” or “SMOOTHING CAPACITOR” or “RESERVOIR CAPACITOR”. Even after using this capacitor a small amount of ripple will remain.

We place the Filter Capacitor at the output of the rectifier the capacitor will charge to the peak voltage during each half cycle then will discharge its stored energy slowly through the load while the rectified voltage drops to zero, thus trying to keep the voltage as constant as possible.

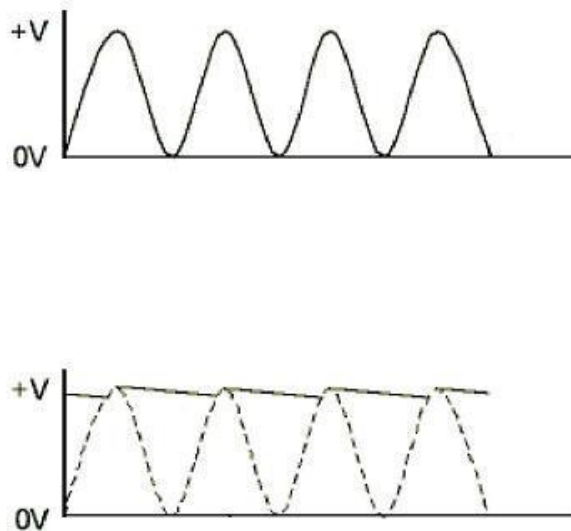


Fig 5.3.1 Filter Capacitor

If we go on increasing the value of the filter capacitor then the Ripple will decrease. But then the costing will increase. The value of the Filter capacitor depends on the current consumed by the circuit, the frequency of the waveform & the accepted ripple.

$$C = \frac{V_r F}{I}$$

Where,

V_r = accepted ripple voltage.(should not be more than 10% of the voltage)

I = current consumed by the circuit in Amperes.

F = frequency of the waveform. A half wave rectifier has only one peak in one cycle so $F=25\text{hz}$

Whereas a full wave rectifier has Two peaks in one cycle so $F=100\text{hz}$.

VOLTAGE REGULATOR

A Voltage regulator is a device which converts varying input voltage into a constant regulated output voltage. Voltage regulator can be of two types

- 1) **Linear Voltage Regulator:** Also called as Resistive Voltage regulator because they dissipate the excessive voltage resistively as heat.
- 2) **Switching Regulators:** They regulate the output voltage by switching the Current ON/OFF very rapidly. Since their output is either ON or OFF it dissipates very low power thus achieving higher efficiency as compared to linear voltage regulators. But they are more complex & generate high noise due to their switching action. For low level of output power switching regulators tend to be costly but for higher output wattage they are much cheaper than linear regulators.

The most commonly available Linear Positive Voltage Regulators are the 78XX series where the XX indicates the output voltage. And 79XX series is for Negative Voltage Regulators.

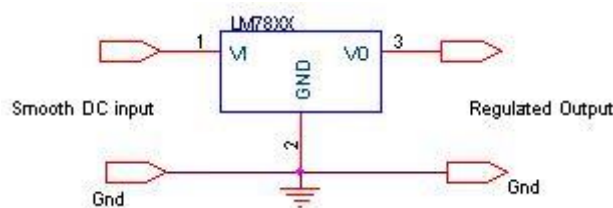


Fig 5.3.2 Voltage regulator

After filtering the rectifier output the signal is given to a voltage regulator. The maximum input voltage that can be applied at the input is 35V. Normally there is a 2-3 Volts drop across the regulator so the input voltage should be at least 2-3 Volts higher than the output voltage. If the input voltage gets below the V_{min} of the regulator due to the ripple voltage or due to any other reason the voltage regulator will not be able to produce the correct regulated voltage.

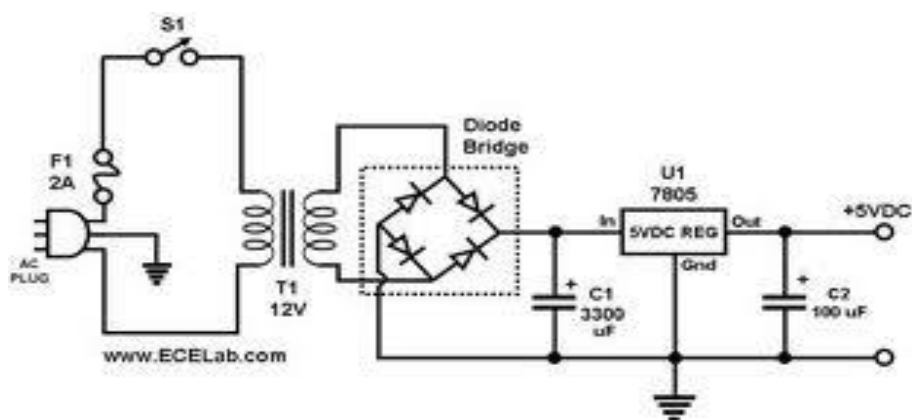


Fig 5.3.3 Circuit Diagram of power supply

IC 7805

7805 is an integrated three-terminal positive fixed linear voltage regulator. It supports an input voltage of 10 volts to 35 volts and output voltage of 5 volts. It has a current rating of 1 amp although lower current models are available. Its output voltage is fixed at 5.0V. The 7805 also has a built-in current limiter as a safety feature. 7805 is manufactured by many companies, including National Semiconductors and Fairchild Semiconductors.

The 7805 will automatically reduce output current if it gets too hot. The last two digits represent the voltage; for instance, the 7812 is a 12-volt regulator. The 78xx series of regulators is designed to work in complement with the 79xx series of negative voltage regulators in systems

that provide both positive and negative regulated voltages, since the 78xx series can't regulate negative voltages in such a system.

The 7805 & 78 is one of the most common and well-known of the 78xx series regulators, as its small component count and medium-power regulated 5V make it useful for powering TTL devices. The IC7805 is a popular voltage regulator integrated circuit commonly used in firefighting robots to regulate the input voltage from sources like batteries or generators to a stable +5 volts DC output. It features built-in thermal overload and short circuit protection, ensuring safe operation in harsh environments.

SPECIFICATIONS	IC 7805
V_{out}	5V
$V_{in} - V_{out}$ Difference	5V - 20V
Operation Ambient Temp	0 - 125°C
Output I_{max}	1A

Table 5.3.1 Specifications of IC7805

LCD DISPLAY

LCD Background:

One of the most common devices attached to a micro controller is an LCD display. Some of the most common LCD's connected to the many microcontrollers are 16x2 and 20x2 displays. This means 16 characters per line by 2 lines and 20 characters per line by 2 lines, respectively.(Basic 16x 2 Characters LCD)

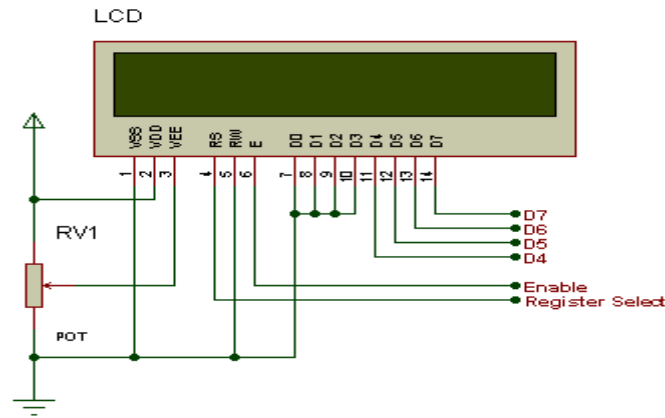


Fig 5.3.4 LCD Pin diagram

Pin No.	Name	Description
Pin no. 1	VSS	Power supply (GND)
Pin no. 2	VCC	Power supply (+5V)
Pin no. 3	VEE	Contrast adjust
Pin no. 4	RS	0 = Instruction input 1 = Data input
Pin no. 5	R/W	0 = Write to LCD module 1 = Read from LCD module
Pin no. 6	EN	Enable signal
Pin no. 7	D0	Data bus line 0 (LSB)
Pin no. 8	D1	Data bus line 1
Pin no. 9	D2	Data bus line 2
Pin no. 10	D3	Data bus line 3
Pin no. 11	D4	Data bus line 4
Pin no. 12	D5	Data bus line 5
Pin no. 13	D6	Data bus line 6
Pin no. 14	D7	Data bus line 7 (MSB)

Table 5.3.2 Character LCD pins with Microcontroller

The LCD requires 3 control lines as well as either 4 or 8 I/O lines for the data bus. The user may select whether the LCD is to operate with a 4-bit data bus or an 8-bit data bus. If a 4-bit data bus is used the LCD will require a total of 7 data lines (3 control lines plus the 4 lines for the data bus). If an 8-bit data bus is used the LCD will require a total of 11 data lines (3 control lines plus the 8 lines for the data bus). The three control lines are referred to as **EN**, **RS**, and **RW**.

The **EN** line is called "Enable." This control line is used to tell the LCD that we are sending it data. To send data to the LCD, our program should make sure this line is low (0) and then set the other two control lines and/or put data on the data bus. When the other lines are completely ready, bring **EN** high (1) and wait for the minimum amount of time required by the LCD datasheet (this varies from LCD to LCD), and end by bringing it low (0) again.

The **RS** line is the "Register Select" line. When RS is low (0), the data is to be treated as a command or special instruction (such as clear screen, position cursor, etc.). When RS is high (1), the data being sent is text data which should be displayed on the screen. For example, to display the letter "T" on the screen we would set RS high.

The **RW** line is the "Read/Write" control line. When RW is low (0), the information on the data bus is being written to the LCD. When RW is high (1), the program is effectively querying (or reading) the LCD. Only one instruction ("Get LCD status") is a read command. All others are write commands--so RW will almost always be low.

Finally, the data bus consists of 4 or 8 lines (depending on the mode of operation selected by the user). In the case of an 8-bit data bus, the lines are referred to as DB0, DB1, DB2, DB3, DB4, DB5, DB6, and DB7.

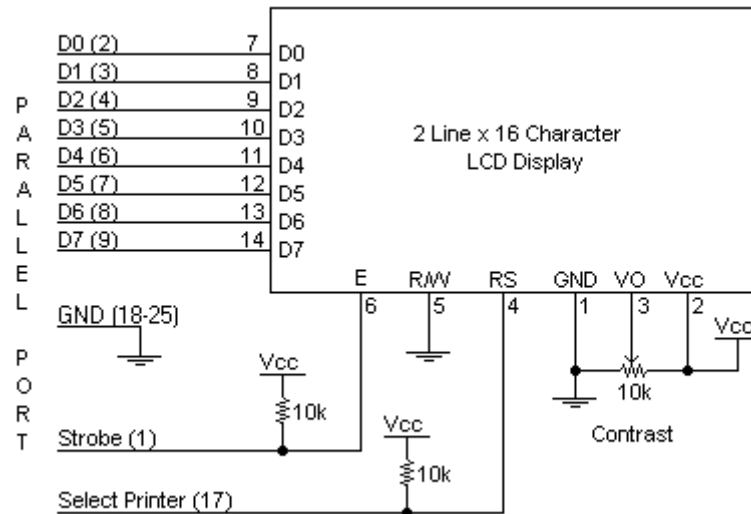


Fig 5.3.5 Circuit Diagram of LCD

Circuit Description

Above is the quite simple schematic. The LCD panel's Enable and Register Select is connected to the Control Port. The Control Port is an open collector / open drain output. While most Parallel Ports have internal pull-up resistors, there is a few which don't. Therefore by incorporating the two 10K external pull up resistors, the circuit is more portable for a wider range of computers, some of which may have no internal pull up resistors.

We make no effort to place the Data bus into reverse direction. Therefore we hard wire the R/W line of the LCD panel, into write mode. This will cause no bus conflicts on the data lines. As a result we cannot read back the LCD's internal Busy Flag which tells us if the LCD has accepted and finished processing the last instruction. This problem is overcome by inserting known delays into our program.

The 10k Potentiometer controls the contrast of the LCD panel. Nothing fancy here. As with all the examples, I've left the power supply out. We can use a bench power supply set to 5v or use an onboard +5 regulator. Remember a few de-coupling capacitors, especially if we have trouble with the circuit working properly.

SETB RW

Handling the EN control line: As we mentioned above, the EN line is used to tell the LCD that we are ready for it to execute an instruction that we've prepared on the data bus and on the other control lines. Note that the EN line must be raised/ lowered before/after each instruction sent to the LCD regardless of whether that instruction is read or write text or instruction. In short, we must always manipulate EN when communicating with the LCD. EN is the LCD's way of knowing that we are talking to it. If we don't raise/lower EN, the LCD doesn't know we're talking to it on the other lines.

Thus, before we interact in any way with the LCD we will always bring the **EN** line low with the following instruction:

CLR EN

And once we've finished setting up our instruction with the other control lines and data bus lines, we'll always bring this line high:

SETB EN

The line must be left high for the amount of time required by the LCD as specified in its datasheet. This is normally on the order of about 250 nanoseconds, but check the datasheet. In the case of a typical microcontroller running at 12 MHz, an instruction requires 1.08 microseconds to execute so the EN line can be brought low the very next instruction. However, faster microcontrollers (such as the DS89C420 which executes an instruction in 90 nanoseconds given an 11.0592 MHz crystal) will require a number of NOPs to create a delay while EN is held high. The number of NOPs that must be inserted depends on the microcontroller we are using and the crystal we have selected.

The instruction is executed by the LCD at the moment the EN line is brought low with a final CLR EN instruction.

Checking the busy status of the LCD:

As previously mentioned, it takes a certain amount of time for each instruction to be executed by the LCD. The delay varies depending on the frequency of the crystal attached to the oscillator input of the LCD as well as the instruction which is being executed.

While it is possible to write code that waits for a specific amount of time to allow the LCD to execute instructions, this method of "waiting" is not very flexible. If the crystal frequency is changed, the software will need to be modified. A more robust method of programming is to use the "Get LCD Status" command to determine whether the LCD is still busy executing the last instruction received.

The "Get LCD Status" command will return to us two tidbits of information; the information that is useful to us right now is found in DB7. In summary, when we issue the "Get LCD Status" command the LCD will immediately raise DB7 if it's still busy executing a command or lower DB7 to indicate that the LCD is no longer occupied. Thus our program can query the LCD until DB7 goes low, indicating the LCD is no longer busy. At that point we are free to continue and send the next command.

Applications:

- Medical equipment
- Electronic test equipment
- Industrial machinery Interface
- Serial terminal
- Advertising system
- Restaurant ordering systems
- Gaming box
- Security systems
- Climatizing units
- PLC Interface
- Simulators

- Environmental monitoring
- Lab development
- Student projects
- Home automation
- PC external display

HC 05 MODULE

The **HC-05** is a very cool module which can add two-way (full-duplex) wireless functionality to your projects. You can use this module to communicate between two microcontrollers like Arduino or communicate with any device with Bluetooth functionality like a Phone or Laptop. There are many android applications that are already available which makes this process a lot easier. The module communicates with the help of USART at 9600 baud rate hence it is easy to interface with any microcontroller that supports USART. We can also configure the default values of the module by using the command mode. So if you looking for a Wireless module that could transfer data from your computer or mobile phone to microcontroller or vice versa then this module might be the right choice for you. However do not expect this module to transfer multimedia like photos or songs; you might have to look into the CSR8645 module for that.

How to Use the HC-05 Bluetooth module

The **HC-05** has two operating modes, one is the Data mode in which it can send and receive data from other Bluetooth devices and the other is the AT Command mode where the default device settings can be changed. We can operate the device in either of these two modes by using the key pin as explained in the pin description.

It is very easy to pair the HC-05 module with microcontrollers because it operates using the Serial Port Protocol (SPP). Simply power the module with +5V and connect the Rx pin of the module to the Tx of MCU and Tx pin of module to Rx of MCU as shown in the figure below

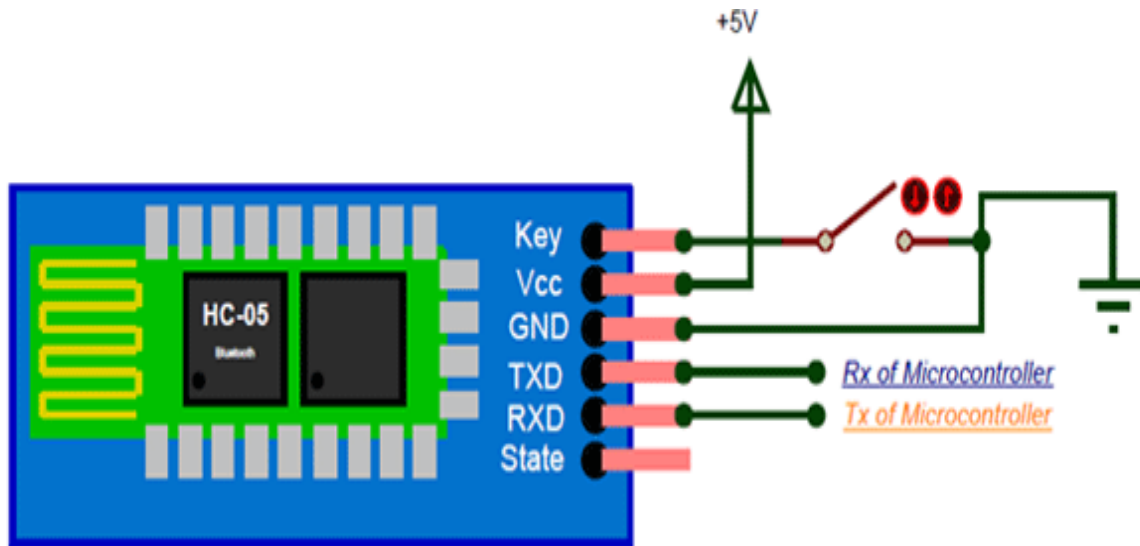


Fig 5.3.6 Bluetooth module

During power up the key pin can be grounded to enter into Command mode, if left free it will by default enter into the data mode. As soon as the module is powered you should be able to discover the Bluetooth device as “HC-05” then connect with it using the default password 1234 and start communicating with it. The name password and other default parameters can be changed by entering into the

Applications

1. Wireless communication between two microcontrollers
2. Communicate with Laptop, Desktops and mobile phones
3. Data Logging application
4. Consumer applications
5. Wireless Robots
6. Home Automation

Pin Number	Pin Name	Description
1	Enable / Key	This pin is used to toggle between Data Mode (set low) and AT command mode (set high). By default it is in Data mode
2	Vcc	Powers the module. Connect to +5V Supply voltage
3	Ground	Ground pin of module, connect to system ground.
4	TX – Transmitter	Transmits Serial Data. Everything received via Bluetooth will be given out by this pin as serial data.
5	RX – Receiver	Receive Serial Data. Every serial data given to this pin will be broadcasted via Bluetooth
6	State	The state pin is connected to on board LED, it can be used as a feedback to check if Bluetooth is working properly.
7	LED	Indicates the status of Module • Blink once in 2 sec: Module has entered Command Mode • Repeated Blinking: Waiting for connection in Data Mode • Blink twice in 1 sec: Connection successful in Data Mode
8	Button	Used to control the Key/Enable pin to toggle between Data and command Mode

Table 5.3.3 Pin Description of Bluetooth Module

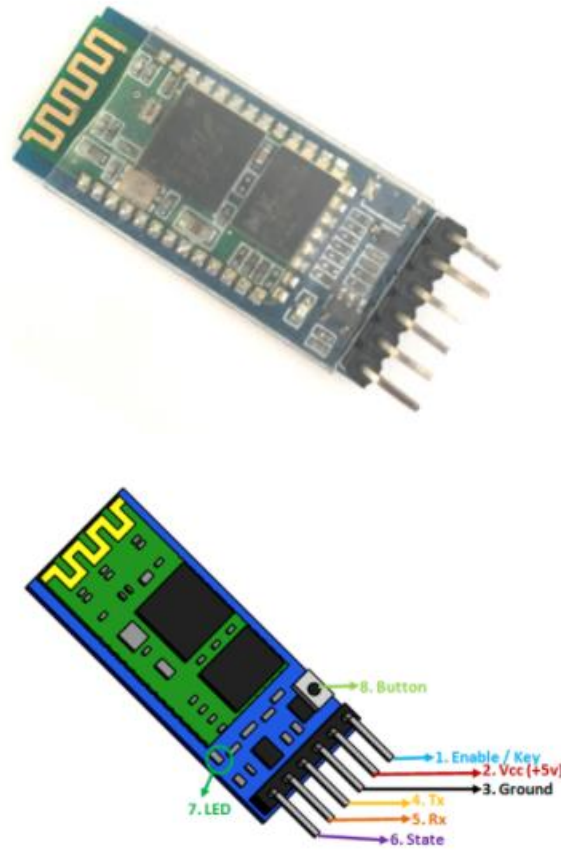


Fig 5.3.7 HC 05 module

FIRE SENSOR

A **flame detector** is a sensor designed to detect and respond to the presence of a flame or fire. Responses to a detected flame depend on the installation, but can include sounding an alarm, deactivating a fuel line (such as a propane or a natural gas line), and activating a fire suppression system.

There are different types of flame detection methods. Some of them are: Ultraviolet detector, near IR array detector, infrared (IR) detector, Infrared thermal cameras, UV/IR detector etc.

When fire burns it emits a small amount of Infra-red light, this light will be received by the Photodiode (IR receiver) on the sensor module. Then we use an Op-Amp to check for change in

voltage across the IR Receiver, so that if a fire is detected the output pin (DO) will give 0V(LOW) and if there is no fire the output pin will be 5V(HIGH)

In this project we are using an **IR based flame sensor**. It is based on the YG1006 sensor which is a high speed and high sensitive NPN silicon phototransistor. It can detect infrared light with a wavelength ranging from 700nm to 1000nm and its detection angle is about 60°. Flame sensor module consists of a photodiode (IR receiver), resistor, capacitor, potentiometer, and LM393 comparator in an integrated circuit. The sensitivity can be adjusted by varying the on board potentiometer. Working voltage is between 3.3v and 5v DC, with a digital output.

Logic high on the output indicates presence of flame or fire. Logic low on output indicates absence of flame or fire.

Below is the Pin Description of Flame sensor Module:

Pin	Description
VCC	3.3 – 5V power supply
Ground	Ground
Dout	Digital output

Table 5.3.4 Pin Description of Flame sensor Module

Once you are ready with your hardware, you can upload the Arduino code for some action. The **complete program** is given at the end of this page. However I have further explained few important bits and pieces here.

As we know the fire sensor will output a HIGH when there is fire and will output a LOW when there is no fire. So we have to keep checking these sensor if any fire has occurred.

GAS SENSOR:

Gas sensors need to be calibrated and periodically checked to ensure sensor accuracy and system integrity. It is important to install stationary sensors in locations where the calibration can be performed easily. The intervals between calibrations can be different from sensor to sensor.

Generally, the manufacturer of the sensor will recommend a time interval between calibrations. However, it is good general practice to check the sensor more closely during the first 30 days after installation. During this period, it is possible to observe how well the sensor is adapting to its new environment. Also, factors that were not accounted for in the design of the system might surface and can affect the sensor's performance.

If the sensor functions properly for 30 continuous days, this provides a good degree of confidence about the installation. Any possible problems can be identified and corrected during this time. Experience indicates that a sensor surviving 30 days after the initial installation will have a good chance of performing its function for the duration expected. Most problems—such as an inappropriate sensor location, interference from other gases, or the loss of sensitivity—will surface during this time.



Fig 5.3.8 Diagram of Gas Sensor

Working of Gas Sensor

During the first 30 days, the sensor should be checked weekly. Afterward, a maintenance schedule, Hazardous Gas Monitors including calibration intervals, should be established. Normally, a monthly calibration is adequate to ensure the effectiveness and sensibility of each sensor; this monthly check will also afford you the opportunity to maintain the system's accuracy. The method and procedure for calibrating the sensors should be established immediately. The calibration procedure should be simple, straightforward, and easily executed by regular personnel. Calibration here is simply a safety check, unlike laboratory analyzers that require a high degree of accuracy. For area air quality and safety gas monitors, the requirements

need to be simple, repeatable, and economical. The procedure should be consistent and traceable. The calibration will be performed in the field where sensors are installed so it can occur in any type environment. Calibration of the gas sensor involves two steps. First the “zero” must be set and then the “span” must be calibrated.

The sensing material in TGS gas sensors is metal oxide, most typically SnO₂. When a metal oxide Crystal such as SnO₂ is heated at a certain high temperature in air, oxygen is adsorbed on the crystal surface with a negative charge. Then donor electrons in the crystal surface are transferred to the adsorbed oxygen, resulting in leaving positive charges in a space charge layer. Thus, surface potential is formed to serve as a potential barrier against electron flow.

Inside the sensor, electric current flows through the conjunction parts (grain boundary) of SnO₂ micro crystals. At grain boundaries, adsorbed oxygen forms a potential barrier which prevents carriers from moving freely. The electrical resistance of the sensor is attributed to this potential barrier. In the presence of a deoxidizing gas, the surface density of the negatively charged oxygen decreases, so the barrier height in the grain boundary is reduced. The reduced barrier height decreases sensor resistance.

Sensor resistance will drop very quickly when exposed to gas, and when removed from gas its resistance will recover to its original value after a short time. The speed of response and reversibility will vary according to the model of sensor and the gas involved.

Feature

- Power requirements: 5 VDC @ ~160mA
- Interface Type: Resistive
- Dimensions: 0.75" diameter x 0.65" tall excluding leads (19.1mm diameter x 16.55mm tall excluding leads)
- Operating temp range: -4 to +122 °F (-20 to +50 °C)

Connections

connecting five volts across the heating (H) pins keeps the sensor hot enough to function correctly. Connecting five volts at either the A or B pins causes the sensor to emit an analog voltage on the other pins. A resistive load between the output pins and ground sets the sensitivity of the detector. Please note that the picture in the datasheet for the top configuration is wrong. Both configurations have the same pinout consistent with the bottom configuration. The resistive load should be calibrated for your particular application using the equations in the datasheet, but a good starting value for the resistor is 10 k Ω .

RELAY

A **relay** is an electrically operated switch. It consists of a set of input terminals for a single or multiple control signals, and a set of operating contact terminals. The switch may have any number of contacts in multiple contact forms, such as make contacts, break contacts, or combinations thereof.

Relays are used where it is necessary to control a circuit by an independent low-power signal, or where several circuits must be controlled by one signal. Relays were first used in long-distance telegraph circuits as signal repeaters: they refresh the signal coming in from one circuit by transmitting it on another circuit. Relays were used extensively in telephone exchanges and early computers to perform logical operations.

The traditional form of a relay uses an electromagnet to close or open the contacts, but other operating principles have been invented, such as in solid-state relays which use semiconductor properties for control without relying on moving parts. Relays with calibrated operating characteristics and sometimes multiple operating coils are used to protect electrical circuits from overload or faults; in modern electric power systems these functions are performed by digital instruments still called protective relays.

Latching relays require only a single pulse of control power to operate the switch persistently. Another pulse applied to a second set of control terminals, or a pulse with opposite polarity, resets the switch, while repeated pulses of the same kind have no effects. Magnetic latching relays are useful in applications when interrupted power should not affect the circuits that the relay is controlling.



Fig 5.3.9 Relay

L293D MOTOR DRIVER WITH MOTOR

D.C. Motor

A dc motor uses electrical energy to produce mechanical energy, very typically through the interaction of magnetic fields and current-carrying conductors. The reverse process, producing electrical energy from the mechanical energy, is accomplished by an alternator, generator or dynamo. Many types of electric motors can be run as generators, and vice versa. The input of a DC motor is current/voltage and its output is torque (speed).



Fig 5.3.10 DC Motor

The DC motor has two basic parts: the rotating part that is called the armature and the stationary part that includes coils of wire called the field coils. The stationary part is also called the stator. Figure shows a picture of a typical DC motor, Figure shows a picture of a DC armature, and Fig shows a picture of a typical stator. From the picture you can see the armature is made of coils of wire wrapped around the core, and the core has an extended shaft that rotates on bearings. You

should also notice that the ends of each coil of wire on the armature are terminated at one end of the armature. The termination points are called the commutator, and this is where the brushes make electrical contact to bring electrical current from the stationary part to the rotating part of the machine.

Operation

The DC motor you will find in modern industrial applications operates very similarly to the simple DC motor described earlier in this chapter. Figure 12-9 shows an electrical diagram of a simple DC motor. Notice that the DC voltage is applied directly to the field winding and the brushes. The armature and the field are both shown as a coil of wire.

In later diagrams, a field resistor will be added in series with the field to control the motor speed. When voltage is applied to the motor, current begins to flow through the field coil from the negative terminal to the positive terminal. This sets up a strong magnetic field in the field winding. Current also begins to flow through the brushes into a commutator segment and then through an armature coil.

The current continues to flow through the coil back to the brush that is attached to other end of the coil and returns to the DC power source. The current flowing in the armature coil sets up a strong magnetic field in the armature.

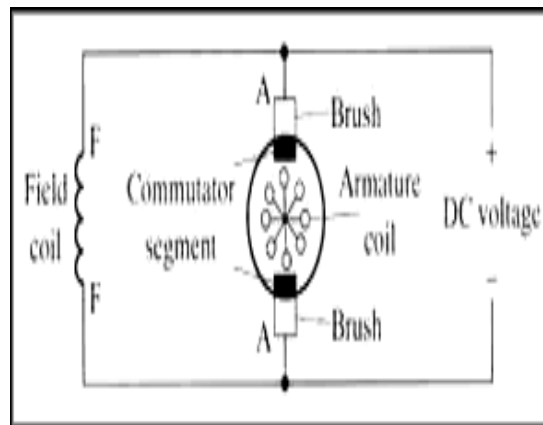


Fig 5.3.11 Simple electrical diagram of DC motor

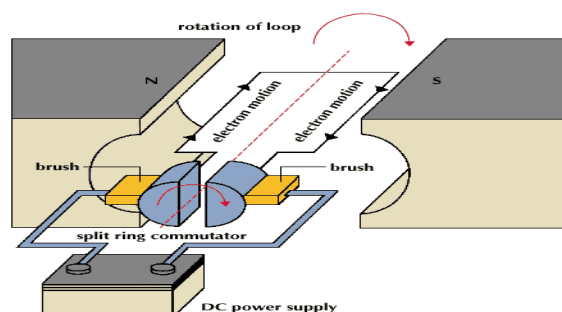


Fig 5.3.12 Operation of a DC Motor

The magnetic field in the armature and field coil causes the armature to begin to rotate. This occurs by the unlike magnetic poles attracting each other and the like magnetic poles repelling each other. As the armature begins to rotate, the commutator segments will also begin to move under the brushes. As an individual commutator segment moves under the brush connected to positive voltage, it will become positive, and when it moves under a brush connected to negative voltage it will become negative. In this way, the commutator segments continually change polarity from positive to negative. Since the commutator segments are connected to the ends of the wires that make up the field winding in the armature, it causes the magnetic field in the armature to change polarity continually from north pole to south pole. The commutator segments and brushes are aligned in such a way that the switch in polarity of the armature coincides with the location of the armature's magnetic field and the field winding's magnetic field. The switching action is timed so that the armature will not lock up magnetically with the field. Instead the magnetic fields tend to build on each other and provide additional torque to keep the motor shaft rotating.

Types of DC motors

1. DC Shunt Motor,
2. DC Series Motor,
3. DC Long Shunt Motor (Compound)
4. DC Short Shunt Motor (Compound)

The rotational energy that you get from any motor is usually the battle between two magnetic fields chasing each other. The DC motor has magnetic poles and an armature, to which DC electricity is fed, The Magnetic Poles are electromagnets, and when they are energized, they produce a strong magnetic field around them, and the armature which is given power with a commutator, constantly repels the poles, and therefore rotates.

1. The DC Shunt Motor

In a 2 pole DC Motor, the armature will have two separate sets of windings, connected to a commutator at the end of the shaft that are in constant touch with carbon brushes. The brushes are static, and the commutator rotate and as the portions of the commutator touching the respective positive or negative polarity brush will energize the respective part of the armature with the respective polarity. It is usually arranged in such a way that the armature and the poles are always repelling.

The general idea of a DC Motor is, the stronger the Field Current, the stronger the magnetic field, and faster the rotation of the armature. When the armature revolves between the poles, the magnetic field of the poles induce power in the armature conductors, and some electricity is generated in the armature, which is called back emf, and it acts as a resistance for the armature. Generally an armature has resistance of less than 1 Ohm, and powering it with heavy voltages of Direct Current could result in immediate short circuits. This back emf helps us there.

When an armature is loaded on a DC Shunt Motor, the speed naturally reduces, and therefore the back emf reduces, which allows more armatures current to flow. This results in more armature field, and therefore it results in torque.

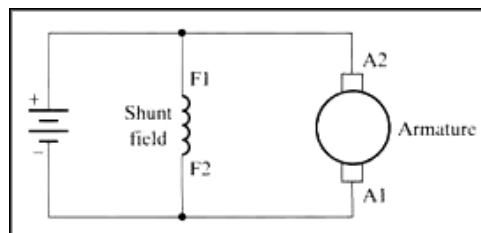


Fig 5.3.13 Diagram of DC shunt motor

When a DC Shunt Motor is overloaded, if the armature becomes too slow, the reduction of the back emf could cause the motor to burn due to heavy current flow thru the armature. The poles and armature are excited separately, and parallel, therefore it is called a Shunt Motor.

2. The DC Series Motor

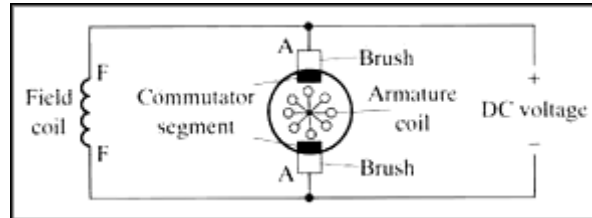


Fig 5.3.14 Diagram of DC series motor

A DC Series Motor has its field coil in series with the armature. Therefore any amount of power drawn by the armature will be passed thru the field. As a result you cannot start a Series DC Motor without any load attached to it. It will either run uncontrollably in full speed, or it will stop.

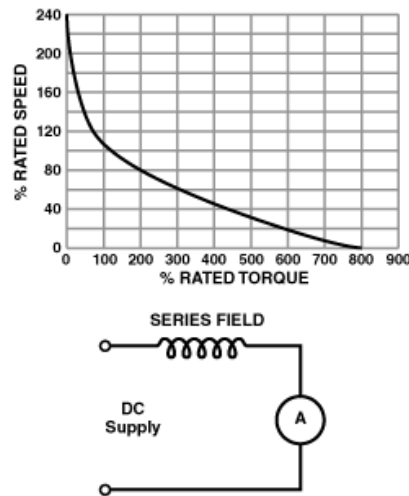


Fig 5.3.15 Diagram of DC series motor graph representation

When the load is increased then its efficiency increases with respect to the load applied. So these are on Electric Trains and elevators.

3. DC Compound Motor:

A compound of Series and Shunt excitation for the fields is done in a Compound DC Motor. This gives the best of both series and shunt motors. Better torque as in a series motor, while the possibility to start the motor with no load.

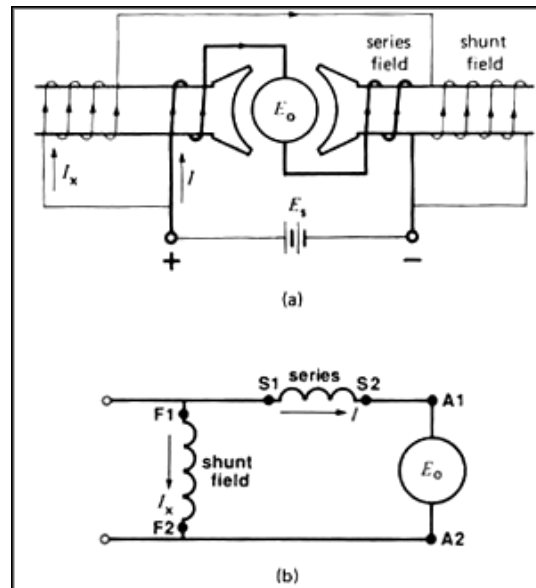


Fig 5.3.16 Diagram of DC compound motor

Above is the diagram of a long shunt motor, while in a short shunt, the shunt coil will be connected after the serial coil. A Compound motor can be run as a shunt motor without connecting the serial coil at all but not vice versa.

DC Motor Driver

The L293 and L293D are quadruple high-current half-H drivers. The L293 is designed to provide bidirectional drive currents of up to 1 A at voltages from 4.5 V to 36 V. The L293D is designed to provide bidirectional drive currents of up to 600-mA at voltages from 4.5 V to 36 V. Both devices are designed to drive inductive loads such as relays, solenoids, dc and

bipolar stepping motors, as well as other high-current/high-voltage loads in positive-supply applications.

All inputs are TTL compatible. Each output is a complete totem-pole drive circuit, with a Darlington transistor sink and a pseudo-Darlington source. Drivers are enabled in pairs, with drivers 1 and 2 enabled by 1,2EN and drivers 3 and 4 enabled by 3,4EN. When an enable input is high, the associated drivers are enabled and their outputs are active and in phase with their inputs.

When the enable input is low, those drivers are disabled and their outputs are off and in the high-impedance state. With the proper data inputs, each pair of drivers forms a full-H (or bridge) reversible drive suitable for solenoid or motor applications. On the L293, external high-speed output clamp diodes should be used for inductive transient suppression. A VCC1 terminal, separate from VCC2, is provided for the logic inputs to minimize device power dissipation. The L293 and L293D are characterized for operation from 0°C to 70°C.

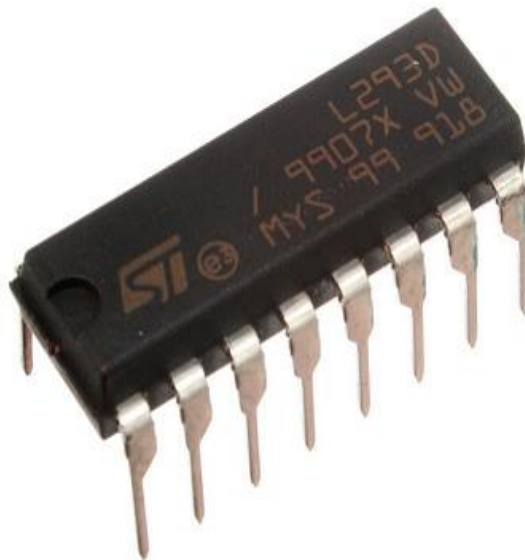


Fig 5.3.17 L293D IC

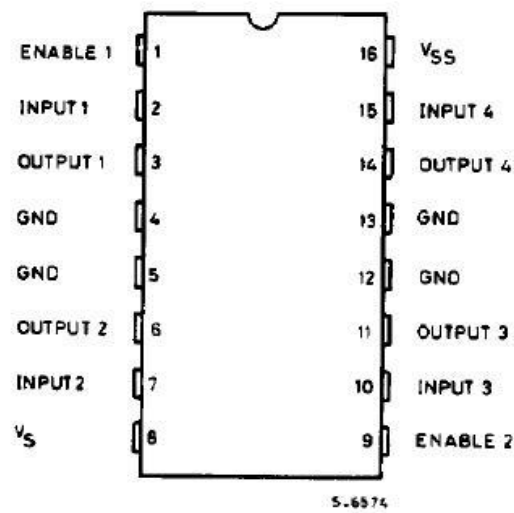


Fig 5.3.18 L293D motor driver pin diagram

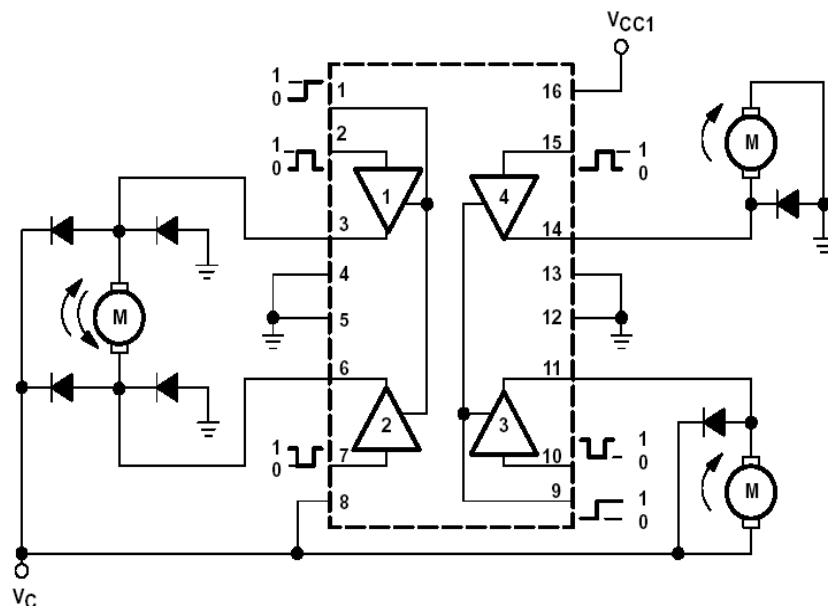


Fig 5.3.19 Internal structure of L293D

Features of L293D

- 600mA Output current capability per channel
- 1.2A Peak output current (non repetitive) per channel
- Enable facility
- Over temperature protection

- Logical “0” input voltage up to 1.5 v
- High noise immunity
- Internal clamp diodes

Applications of DC Motors

1. Electric Train: A kind of DC motor called the DC Series Motor is used in Electric Trains. The DC Series Motors have the property to deliver more power when they are loaded more. So the more the people get on a train, the more powerful the train becomes.
2. Elevators: The best bidirectional motors are DC motors. They are used in elevators. Compound DC Motors are used for this application.
3. PC Fans, CD ROM Drives, and Hard Drives: All these things need motors, very miniature motors, with great precision. AC motors can never imagine any application in these places.
4. Starter Motors in Automobiles: An automobile battery supplies DC, so a DC motor is best suited here. Also, you cannot start an engine with a small sized AC motor.
5. Propulsion: DC motors drive the wheels or tracks of the firefighting robot, enabling mobility over various terrains encountered during firefighting missions.
6. Maneuverability: These motors provide precise control for navigating tight spaces, avoiding obstacles, and adjusting speed and direction as needed.
7. Electrical Machines Lab in Colleges.

CHAPTER 6

ARDUINO CONTROLLER

Arduino is an open-source hardware and software company, project and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices and interactive objects that can sense and control both physically and digitally. Its products are licensed under the GNU Lesser General Public License (LGPL) or the GNU General Public License (GPL), permitting the manufacture of Arduino boards and software distribution by anyone. Arduino boards are available commercially in preassembled form or as do-it-yourself (DIY) kits.

Arduino board designs use a variety of microprocessors and controllers. The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards or breadboards (shields) and other circuits. The boards feature serial communications interfaces, including Universal Serial Bus (USB) on some models, which are also used for loading programs from personal computers.

The microcontrollers are typically programmed using a dialect of features from the programming languages C and C++. In addition to using traditional compiler toolchains, the Arduino project provides an integrated development environment (IDE) based on the Processing language project.

The Arduino project started in 2003 as a program for students at the Interaction Design Institute Ivrea in Ivrea, Italy, aiming to provide a low-cost and easy way for novices and professionals to create devices that interact with their environment using sensors and actuators. Common examples of such devices intended for beginner hobbyists include simple robots, thermostats and motion detectors.

The name Arduino comes from a bar in Ivrea, Italy, where some of the founders of the project used to meet. The bar was named after Arduin of Ivrea, who was the margrave of the March of Ivrea and King of Italy from 1002 to 1014.



Fig 6.1 Hardware image

History

The Arduino project was started at the Interaction Design Institute Ivrea (IDII) in Ivrea, Italy.^[2] At that time, the students used a BASIC Stamp microcontroller at a cost of \$50, a considerable expense for many students. In 2003 Hernando Barragán created the development platform Wiring as a Master's thesis project at IDII, under the supervision of Massimo Banzi and Casey Reas. Casey Reas is known for co-creating, with Ben Fry, the Processing development platform. The project goal was to create simple, low cost tools for creating digital projects by non-engineers. The Wiring platform consisted of a printed circuit board (PCB) with an ATmega168 microcontroller, an IDE based on Processing and library functions to easily program the microcontroller.^[4] In 2003, Massimo Banzi, with David Mellis, another IDII student, and David Cuartielles, added support for the cheaper ATmega8 microcontroller to Wiring. But instead of continuing the work on Wiring, they forked the project and renamed it Arduino.

The initial Arduino core team consisted of Massimo Banzi, David Cuartielles, Tom Igoe, Gianluca Martino, and David Mellis,^[2] but Barragán was not invited to participate.

Following the completion of the Wiring platform, lighter and less expensive versions were distributed in the open-source community.

It was estimated in mid-2011 that over 300,000 official Arduinos had been commercially produced, and in 2013 that 700,000 official boards were in users' hands. In October 2016, Federico Musto, Arduino's former CEO, secured a 50% ownership of the company. In April

2017, Wired reported that Musto had "fabricated his academic record.... On his company's website, personal LinkedIn accounts, and even on Italian business documents, Musto was until recently listed as holding a PhD from the Massachusetts Institute of Technology. In some cases, his biography also claimed an MBA from New York University." Wired reported that neither University had any record of Musto's attendance, and Musto later admitted in an interview with Wired that he had never earned those degrees.

Around that same time, Massimo Banzi announced that the Arduino Foundation would be "a new beginning for Arduino." But a year later, the Foundation still hasn't been established, and the state of the project remains unclear. The controversy surrounding Musto continued when, in July 2017, he reportedly pulled many Open source licenses, schematics, and code from the Arduino website, prompting scrutiny and outcry. In October 2017, Arduino announced its partnership with ARM Holdings (ARM). The announcement said, in part, "ARM recognized independence as a core value of Arduino ... without any lock-in with the ARM architecture." Arduino intends to continue to work with all technology vendors and architectures.

OPERATION WITH PINS

Arduino is open-source hardware. The hardware reference designs are distributed under a Creative Commons Attribution Share-Alike 2.5 license and are available on the Arduino website. Layout and production files for some versions of the hardware are also available.

Although the hardware and software designs are freely available under copyleft licenses, the developers have requested the name Arduino to be exclusive to the official product and not be used for derived works without permission.

The official policy document on use of the Arduino name emphasizes that the project is open to incorporating work by others into the official product. Several Arduino-compatible products commercially released have avoided the project name by using various names ending in -duino.

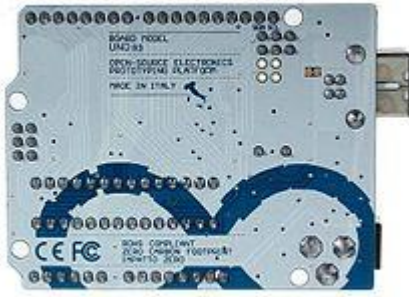


Fig.6.2 Back side of module

Most Arduino boards consist of an Atmel 8-bit AVR microcontroller (ATmega8,[24] ATmega168, ATmega328, ATmega1280, ATmega2560) with varying amounts of flash memory, pins, and features. The 32-bit Arduino Due, based on the Atmel SAM3X8E was introduced in 2012. The boards use single or double-row pins or female headers that facilitate connections for programming and incorporation into other circuits. These may connect with add-on modules termed shields. Multiple and possibly stacked shields may be individually addressable via an I²C serial bus. Most boards include a 5 V linear regulator and a 16 MHz crystal oscillator or ceramic resonator. Some designs, such as the LilyPad, run at 8 MHz and dispense with the onboard voltage regulator due to specific form-factor restrictions.

Arduino microcontrollers are pre-programmed with a boot loader that simplifies uploading of programs to the on-chip flash memory. The default bootloader of the Arduino UNO is the optiboot bootloader. Boards are loaded with program code via a serial connection to another computer. Some serial Arduino boards contain a level shifter circuit to convert between RS-232 logic levels and transistor–transistor logic (TTL) level signals. Current Arduino boards are programmed via Universal Serial Bus (USB), implemented using USB-to-serial adapter chips such as the FTDI FT232. Some boards, such as later-model Uno boards, substitute the FTDI chip with a separate AVR chip containing USB-to-serial firmware, which is reprogrammable via its own ICSP header. Other variants, such as the Arduino Mini and the unofficial Boarduino, use a detachable USB-to-serial adapter board or cable, Bluetooth or other methods. When used with traditional microcontroller tools, instead of the Arduino IDE, standard AVR in-system programming (ISP) programming is used. The Arduino board exposes most of the

microcontroller's I/O pins for use by other circuits. The Diecimila,[a] Duemilanove,[b] and current Uno[c] provide 14 digital I/O pins, six of which can produce pulse-width modulated signals, and six analog inputs, which can also be used as six digital I/O pins. These pins are on the top of the board, via female 0.1-inch (2.54 mm) headers. Several plug-in application shields are also commercially available. The Arduino Nano, and Arduino-compatible Bare Bones Board and Boarduino boards may provide male header pins on the underside of the board that can plug into solderless breadboards.

Many Arduino-compatible and Arduino-derived boards exist. Some are functionally equivalent to an Arduino and can be used interchangeably. Many enhance the basic Arduino by adding output drivers, often for use in school-level education, to simplify making buggies and small robots. Others are electrically equivalent but change the form factor, sometimes retaining compatibility with shields, sometimes not. Some variants use different processors, of varying compatibility.



Fig.6.3 Ardino board

1	Power USB Arduino board can be powered by using the USB cable from your computer. All you need to do is connect the USB cable to the USB connection (1).
2	Power (Barrel Jack) Arduino boards can be powered directly from the AC mains power supply by connecting it to the Barrel Jack (2).

3	Voltage Regulator The function of the voltage regulator is to control the voltage given to the Arduino board and stabilize the DC voltages used by the processor and other elements.
4	Crystal Oscillator The crystal oscillator helps Arduino in dealing with time issues. How does Arduino calculate time? The answer is, by using the crystal oscillator. The number printed on top of the Arduino crystal is 16.000H9H. It tells us that the frequency is 16,000,000 Hertz or 16 MHz.
5,17	Arduino Reset You can reset your Arduino board, i.e., start your program from the beginning. You can reset the UNO board in two ways. First, by using the reset button (17) on the board. Second, you can connect an external reset button to the Arduino pin labelled RESET (5).
6,7,8,9	Pins (3.3, 5, GND, Vin) • 3.3V (6) – Supply 3.3 output volt • 5V (7) – Supply 5 output volt • Most of the components used with Arduino board works fine with 3.3 volt and 5 volt. • GND (8)(Ground) – There are several GND pins on the Arduino, any of which can be used to ground your circuit. • Vin (9) – This pin also can be used to power the Arduino board from an external power source, like AC mains power supply.
10	Analog pins The Arduino UNO board has six analog input pins A0 through A5. These pins can read signal from an analog sensor like humidity sensor or temperature sensor and convert it into a digital value that can be read by microprocessor.

11	Main microcontroller Each Arduino board has its own microcontroller (11). You can assume it as the brain of your board. The main IC (integrated circuit) on the Arduino is slightly different from board to board. The microcontrollers are usually of the ATMEL Company. You must know what IC your board has before loading up a new program from the Arduino IDE. This information is available on the top of the IC. For more details about the IC construction and functions, you can refer to the data sheet.
12	ICSP pin Mostly, ICSP (12) is an AVR, a tiny programming header for the Arduino consisting of MOSI, MISO, SCK, RESET, VCC, and GND. It is often referred to as an SPI (Serial Peripheral Interface), which could be considered as an "expansion" of the output. Actually, you are slaving the output device to the master of the SPI bus.
13	Power LED indicator This LED should light up when you plug your Arduino into a power source to indicate that your board is powered up correctly. If this light does not turn on, then there is something wrong with the connection.
14	TX and RX LEDs On your board, you will find two labels: TX (transmit) and RX (receive). They appear in two places on the Arduino UNO board. First, at the digital pins 0 and 1, to indicate the pins responsible for serial communication. Second, the TX and RX led (13). The TX led flashes with different speed while sending the serial data. The speed of flashing depends on the baud rate used by the board. RX flashes during the receiving process.
15	Digital I/O The Arduino UNO board has 14 digital I/O pins (15) (of which 6 provide PWM (Pulse Width Modulation) output. These pins can be configured to work as input digital pins to read logic values (0 or 1) or as digital output pins to

	drive different modules like LEDs, relays, etc. The pins labeled “~” can be used to generate PWM.
16	AREF AREF stands for Analog Reference. It is sometimes, used to set an external reference voltage (between 0 and 5 Volts) as the upper limit for the analog input pins.

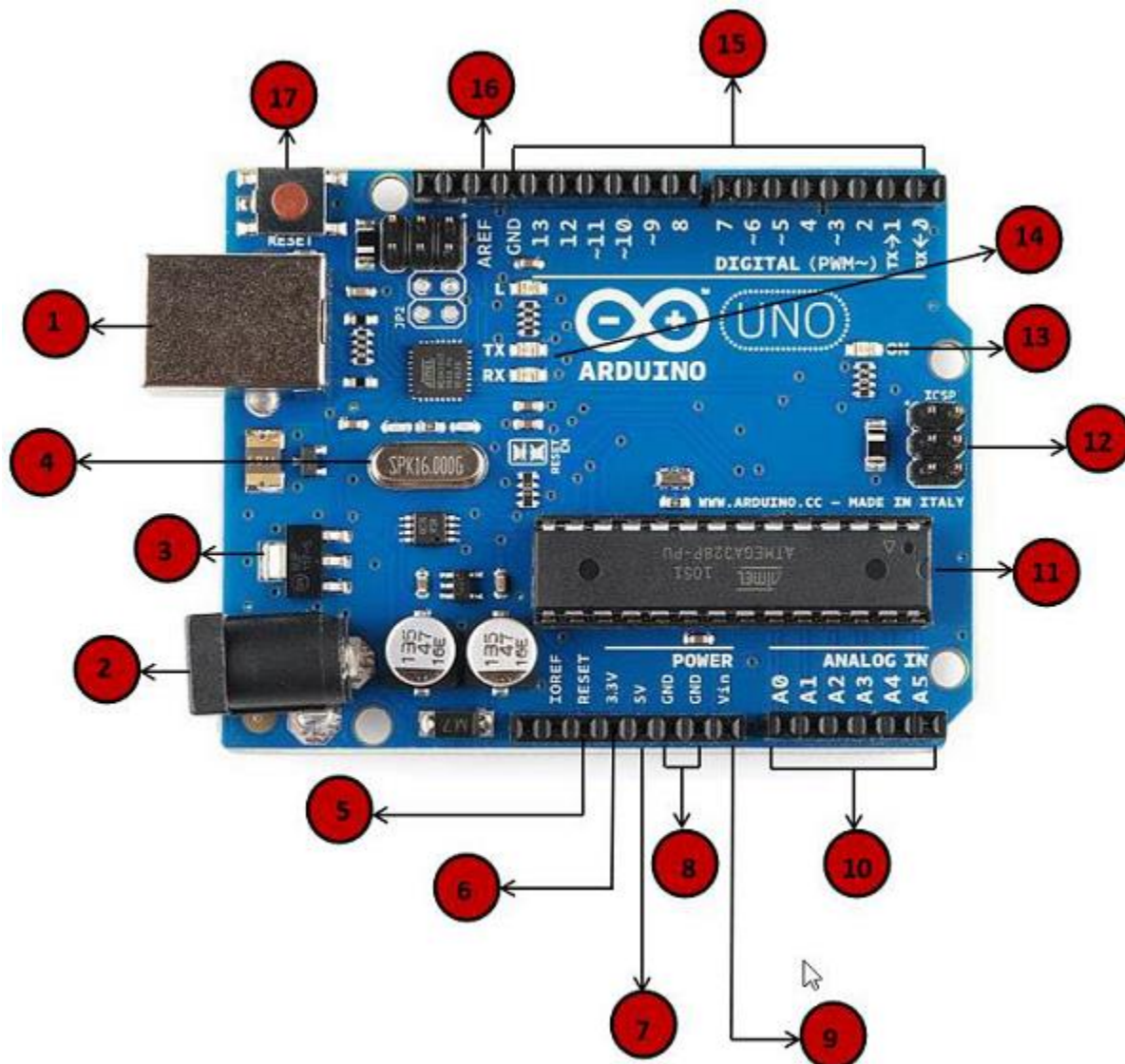


Fig 6.4 Pin explanation

CHAPTER 7

IMPLEMENTATION

7.1 SOFTWARE REQUIREMENTS

This project is implemented using following software's:

- Express PCB – for designing circuit
- Arduino IDE compiler - for compilation part
- Proteus 7 (Embedded C) – for simulation part

7.2 THE INTERFACE

When a project is first started you will be greeted with a yellow outline. This yellow outline is the dimension of the PCB. Typically after positioning of parts and traces, move them to their final position and then crop the PCB to the correct size. However, in designing a board with a certain size constraint, crop the PCB to the correct size before starting. Figure 7.2.1 show the toolbar in which the each button has the following functions.



Fig: 7.2.1 Tool bar necessary for the interface

The select tool: It is fairly obvious what this does. It allows you to move and manipulate parts. When this tool is selected the top toolbar will show buttons to move traces to the top / bottom copper layer, and rotate buttons.

The zoom to selection tool: The zoom to selection tool is used for magnifying a selected part of the image.

The place pad: button allows you to place small soldier pads which are useful for board connections or if a part is not in the part library but the part dimensions are available. When this

tool is selected the top toolbar will give you a large selection of round holes, square holes and surface mount pads.

The place component: tool allows you to select a component from the top toolbar and then by clicking in the workspace places that component in the orientation chosen using the buttons next to the component list. The components can always be rotated afterwards with the select tool if the orientation is wrong.

The place trace: tool allows you to place a solid trace on the board of varying thicknesses. The top toolbar allows you to select the top or bottom layer to place the trace on.

The Insert Corner in trace: button does exactly what it says. When this tool is selected, clicking on a trace will insert a corner which can be moved to route around components and other traces.

The remove a trace button is not very important since the delete key will achieve the same result.

7.3 DESIGN CONSIDERATIONS

Before starting a project there are several ways to design a PCB and one must be chosen to suit the project's needs. Single sided, or double sided?

When making a PCB you have the option of making a single sided board, or a double sided board. Single sided boards are cheaper to produce and easier to etch, but much harder to design for large projects. If a lot of parts are being used in a small space it may be difficult to make a single sided board without jumpering over traces with a cable. While there's technically nothing wrong with this, it should be avoided if the signal travelling over the traces is sensitive (e.g. audio signals).

A double sided board is more expensive to produce professionally, more difficult to etch on a DIY board, but makes the layout of components a lot smaller and easier. It should be noted that if a trace is running on the top layer, check with the components to make sure you can get to its

pins with a soldering iron. Large capacitors, relays, and similar parts which don't have axial leads can NOT have traces on top unless boards are plated professionally.

7.4 CODE IMPLEMENTATION

```
#include <LiquidCrystal.h>
```

```
#include <stdio.h>
```

```
LiquidCrystal lcd(6, 7, 5, 4, 3, 2);
```

```
unsigned char rcv,count,gchr,gchr1,robos='s',modes='x',moment='x';
```

```
int cntlk=0,rty=0;
```

```
//char pastnumber[11]="";
```

```
float tempc=0;
```

```
int sti=0;
```

```
String inputString = "";    // a string to hold incoming data
```

```
boolean stringComplete = false; // whether the string is complete
```

```
int m1a = 8;
```

```
int m1b = 9;
```

```
int m2a = 10;
```

```
int m2b = 11;
```

```
int fire  = 13;

int gas   = 12;

int buzzer = A1;


int relay = A0;


int val1 = 0, val2 = 0;

int dist1, dist2, dist3, sts1=0;

long duration;

int distanceCm, distanceInch;


void okcheck()

{

    unsigned char rcr;

    do{

        rcr = Serial.read();

    }while(rcr != 'K');

}

void setup()

{

    // initialize serial:

    Serial.begin(9600); serialEvent();

    //  serialEvent();
```

```
pinMode(m1a, OUTPUT); pinMode(m1b, OUTPUT);
```

```
pinMode(m2a, OUTPUT); pinMode(m2b, OUTPUT);
```

```
pinMode(relay, OUTPUT);
```

```
pinMode(buzzer, OUTPUT);
```

```
pinMode(fire, INPUT);
```

```
pinMode(gas, INPUT);
```

```
digitalWrite(m1a, LOW); digitalWrite(m1b, LOW);
```

```
digitalWrite(m2a, LOW); digitalWrite(m2b, LOW);
```

```
digitalWrite(relay, LOW);
```

```
digitalWrite(buzzer, HIGH);
```

```
lcd.begin(16, 2);
```

```
lcd.print("Fire Fighting Robot");
```

```
/*
```

```
Serial.write("AT\r\n");    delay(3000); //okcheck();
```

```
Serial.write("ATE0\r\n");    okcheck();
```

```
Serial.write("AT+CWMODE=3\r\n"); delay(2000);
```

```
Serial.write("AT+CIPMUX=1\r\n"); delay(2000); //    okcheck();
```

```
Serial.write("AT+CIPSERVER=1,23\r\n"); //    okcheck();
```

```
lcd.clear();
```

```
lcd.print("Waiting For");
```

```
lcd.setCursor(0,1);
```

```
lcd.print("Connection");
```

```
do{

    rcv = Serial.read();

    }while(rcv != 'C');


    lcd.clear();

    lcd.print("Connected");

    delay(1000);

    */

    delay(2000);


    lcd.clear();

    lcd.setCursor(0, 0);

    lcd.print("Fire:"); //5,0

    lcd.setCursor(0, 1);

    lcd.print("Gas:"); //4,1


    serialEvent();

}


void loop()

{

    if(digitalRead(fire) == LOW)

    {

        lcd.setCursor(5,0);
```

```
    lcd.print("ON ");

    digitalWrite(m1a, LOW); digitalWrite(m1b, LOW);
    digitalWrite(m2a, LOW); digitalWrite(m2b, LOW);

    digitalWrite(relay,HIGH);
    digitalWrite(buzzer, LOW);
    Serial.write("Fire ON\r\n");
}
if(digitalRead(fire) == HIGH)
{
    lcd.setCursor(5,0);
    lcd.print("OFF");
}

if(digitalRead(gas) == LOW)
{
    lcd.setCursor(5,1);
    lcd.print("ON ");

    digitalWrite(m1a, LOW); digitalWrite(m1b, LOW);
    digitalWrite(m2a, LOW); digitalWrite(m2b, LOW);

    digitalWrite(relay,HIGH);
```



```
digitalWrite(buzzer, LOW);

Serial.write("Gas ON\r\n");

}

if(digitalRead(gas) == HIGH)

{

    lcd.setCursor(5,1);

    lcd.print("OFF");

}


if(digitalRead(fire) == HIGH && digitalRead(gas) == HIGH)

{

    digitalWrite(relay,LOW);

    digitalWrite(buzzer, HIGH);

}


/*

if(stringComplete)

{

    if(gchr == 'f' || gchr == 'F')

    {gchr ='x';

    robos='m';

    digitalWrite(m1a, HIGH); digitalWrite(m1b, LOW);

    digitalWrite(m2a, HIGH); digitalWrite(m2b, LOW);
```

```
}
```

```
if(gchr == 'b' || gchr == 'B')
```

```
{gchr ='x';
```

```
  robos='m';
```

```
  digitalWrite(m1a, LOW); digitalWrite(m1b, HIGH);
```

```
  digitalWrite(m2a, LOW); digitalWrite(m2b, HIGH);
```

```
}
```

```
if(gchr == 'l' || gchr == 'L')
```

```
{gchr ='x';
```

```
  robos='m';
```

```
  digitalWrite(m1a, HIGH); digitalWrite(m1b, LOW);
```

```
  digitalWrite(m2a, LOW); digitalWrite(m2b, HIGH);
```

```
}
```

```
if(gchr == 'r' || gchr == 'R')
```

```
{gchr ='x';
```

```
  robos='m';
```

```
  digitalWrite(m1a, LOW); digitalWrite(m1b, HIGH);
```

```
  digitalWrite(m2a, HIGH); digitalWrite(m2b, LOW);
```

```
}
```

```
if(gchr == 's' || gchr == 'S')
```

```
{gchr ='x';  
  
  robos='s';  
  
  digitalWrite(m1a, LOW); digitalWrite(m1b, LOW);  
  
  digitalWrite(m2a, LOW); digitalWrite(m2b, LOW);  
  
}
```

```
if(gchr == '1')  
  
  {gchr ='x';  
  
    digitalWrite(m3a, HIGH); digitalWrite(m3b, LOW);  
  
  }
```

```
if(gchr == '2')  
  
  {gchr ='x';  
  
    digitalWrite(m3a, LOW); digitalWrite(m3b, HIGH);  
  
  }
```

```
if(gchr == '3')  
  
  {gchr ='x';  
  
    digitalWrite(m3a, LOW); digitalWrite(m3b, LOW);  
  
  }
```

```
if(gchr == '4' && modes == 'm')  
  
  {gchr ='x';  
  
    digitalWrite(m4a, HIGH); digitalWrite(m4b, LOW);  
  
  }
```

```
if(gchr == '5' && modes == 'm')
{
    gchr ='x';

    digitalWrite(m4a, LOW); digitalWrite(m4b, HIGH);

}

if(gchr == '6' && modes == 'm')
{
    gchr ='x';

    digitalWrite(m4a, LOW); digitalWrite(m4b, LOW);

}


inputString = "";
stringComplete = false;
}*/
}

void serialEvent()
{
    while (Serial.available() < 0)
    {
        char inChar = (char)Serial.read();

        gchr = inChar;

        if(gchr == 'f' || gchr == 'F')
        {
            gchr ='x';
```

```
    robos='m';

    digitalWrite(m1a, HIGH); digitalWrite(m1b, LOW);

    digitalWrite(m2a, HIGH); digitalWrite(m2b, LOW);
}

if(gchr == 'b' || gchr == 'B')
{
    gchr='x';

    robos='m';

    digitalWrite(m1a, LOW); digitalWrite(m1b, HIGH);

    digitalWrite(m2a, LOW); digitalWrite(m2b, HIGH);
}

if(gchr == 'l' || gchr == 'L')
{
    gchr='x';

    robos='m';

    digitalWrite(m1a, HIGH); digitalWrite(m1b, LOW);

    digitalWrite(m2a, LOW); digitalWrite(m2b, HIGH);
}

if(gchr == 'r' || gchr == 'R')
{
    gchr='x';

    robos='m';

    digitalWrite(m1a, LOW); digitalWrite(m1b, HIGH);

    digitalWrite(m2a, HIGH); digitalWrite(m2b, LOW);
}
```

```
    }

    if(gchr == 's' || gchr == 'S')
    {
        gchr ='x';

        robos='s';

        digitalWrite(m1a, LOW); digitalWrite(m1b, LOW);

        digitalWrite(m2a, LOW); digitalWrite(m2b, LOW);

    }

    inputString = "";

}

/*

sensorValue = analogRead(analogInPin);

sensorValue = (sensorValue/9.31);

lcd.setCursor(1,1); //rc

lcd.print(sensorValue);

Serial.print(sensorValue);

*/

void converts(unsigned int value)
{

    unsigned int a,b,c,d,e,f,g,h;

    a=value/10000;
```

```
b=value%10000;

c=b/1000;

d=b%1000;

e=d/100;

f=d%100;

g=f/10;

h=f%10;


a=a|0x30;

c=c|0x30;

e=e|0x30;

g=g|0x30;

h=h|0x30;


Serial.write(a);

Serial.write(c);

Serial.write(e);

Serial.write(g);

Serial.write(h);

}


void convertl(unsigned int value)

{

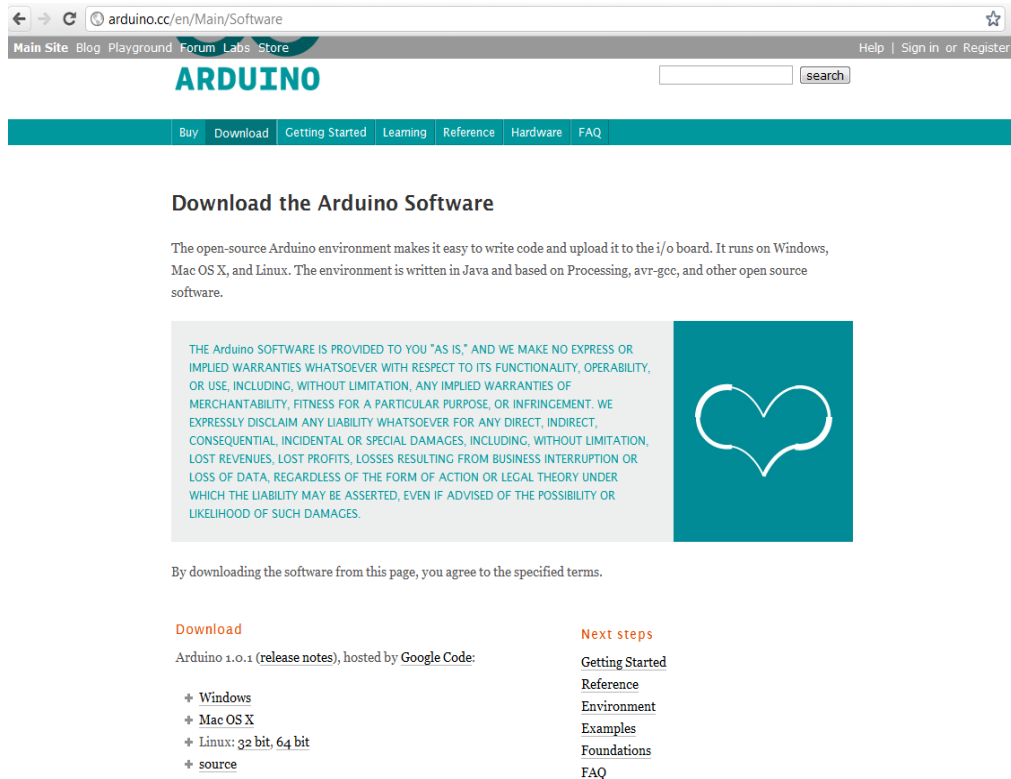
    unsigned int a,b,c,d,e,f,g,h;
```

```
a=value/10000;
b=value%10000;
c=b/1000;
d=b%1000;
e=d/100;
f=d%100;
g=f/10;
h=f%10;

a=a|0x30;
c=c|0x30;
e=e|0x30;
g=g|0x30;
h=h|0x30;

// lcd.write(a);
// lcd.write(c);
lcd.write(e);
lcd.write(g);
lcd.write(h);
}
```


7.5 AURDINO COMPILING



The screenshot shows the Arduino.cc website. The browser address bar displays 'arduino.cc/en/Main/Software'. The website header includes navigation links: 'Main Site', 'Blog', 'Playground', 'Forum', 'Labs', and 'Store'. The Arduino logo is prominently displayed. Below the logo is a search bar. A teal navigation bar contains links: 'Buy', 'Download', 'Getting Started', 'Learning', 'Reference', 'Hardware', and 'FAQ'. The main content area is titled 'Download the Arduino Software'. It contains a paragraph about the open-source environment. Below this is a disclaimer box with a heart icon. A note states that downloading the software implies agreement to the terms. Two columns of links are provided: 'Download' (with links for Windows, Mac OS X, Linux, and source) and 'Next steps' (with links for Getting Started, Reference, Environment, Examples, Foundations, and FAQ).

arduino.cc/en/Main/Software

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ARDUINO

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Download the Arduino Software

The open-source Arduino environment makes it easy to write code and upload it to the i/o board. It runs on Windows, Mac OS X, and Linux. The environment is written in Java and based on Processing, avr-gcc, and other open source software.

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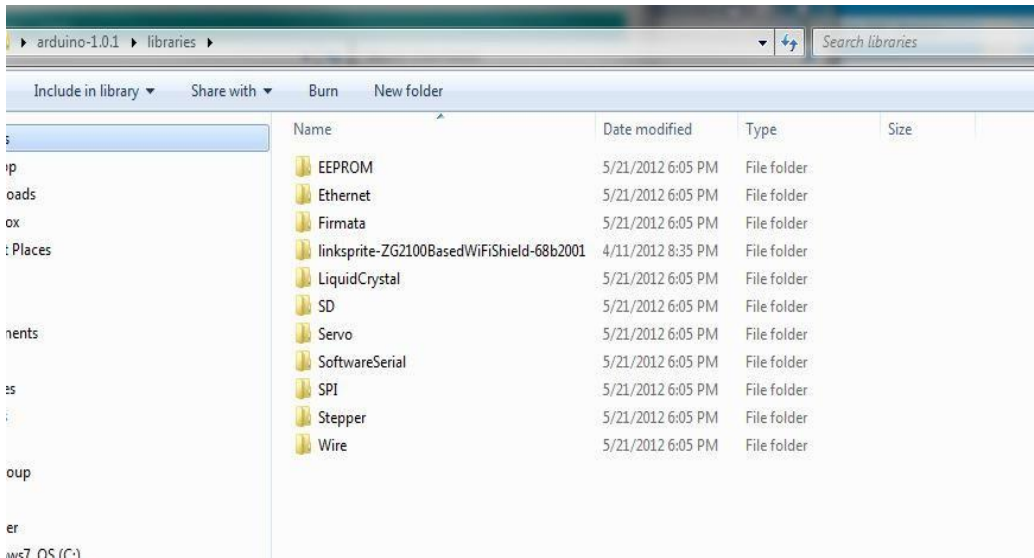
Arduino 1.0.1 (release notes), hosted by Google Code:

- + Windows
- + Mac OS X
- + Linux: 32 bit, 64 bit
- + source

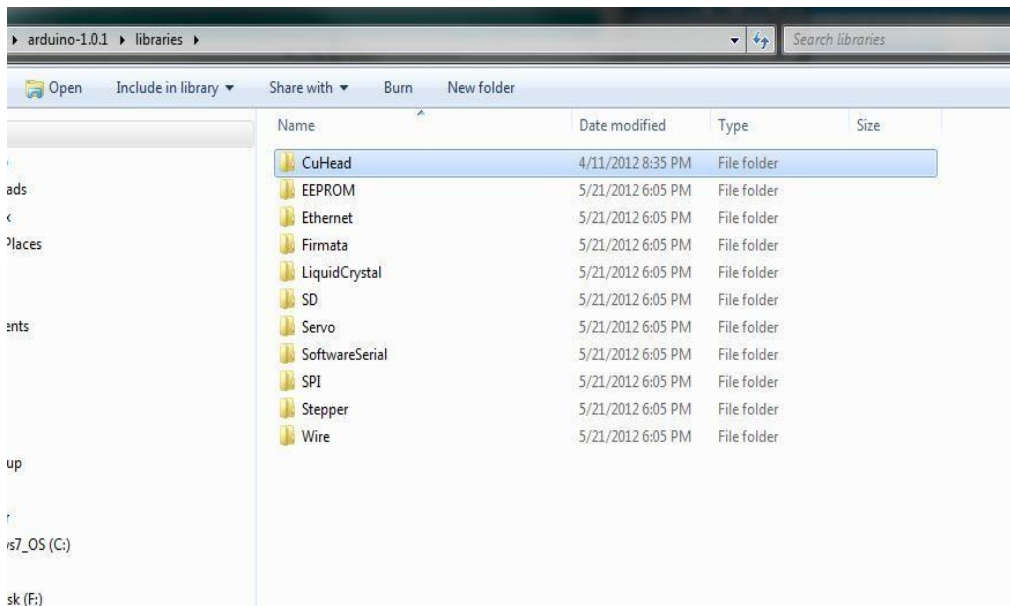
Next steps

- Getting Started
- Reference
- Environment
- Examples
- Foundations
- FAQ

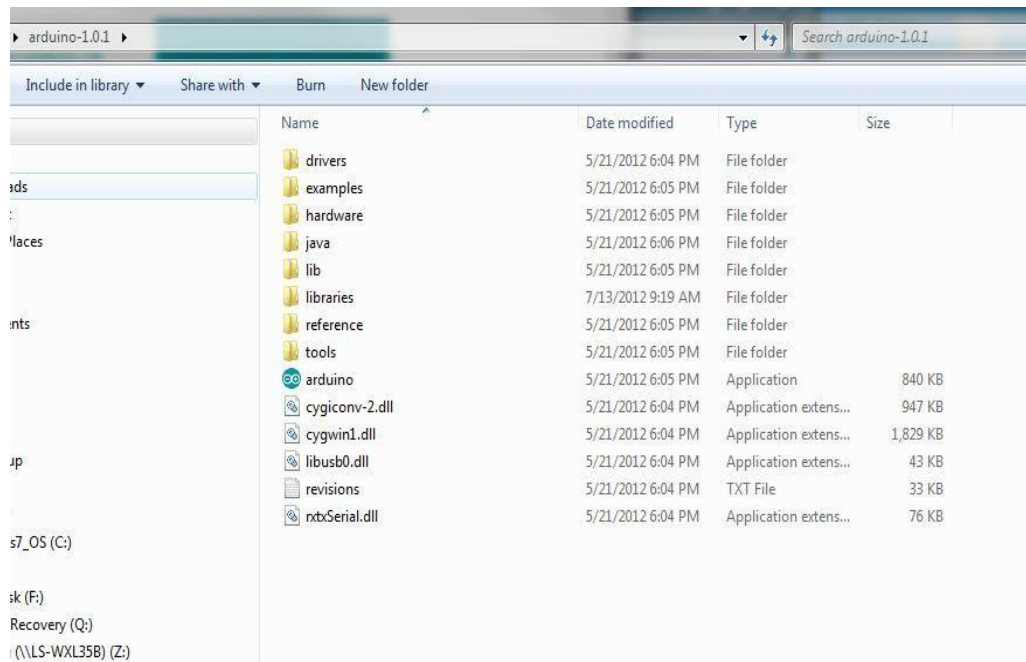
In next step download library



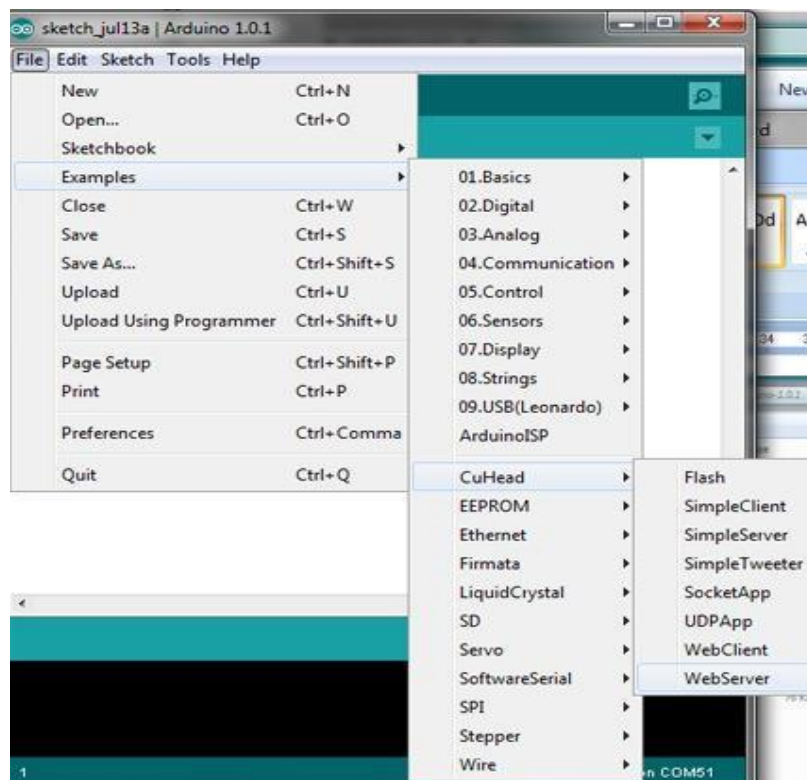
As Arduino doesn't recognize the directory name, please rename it



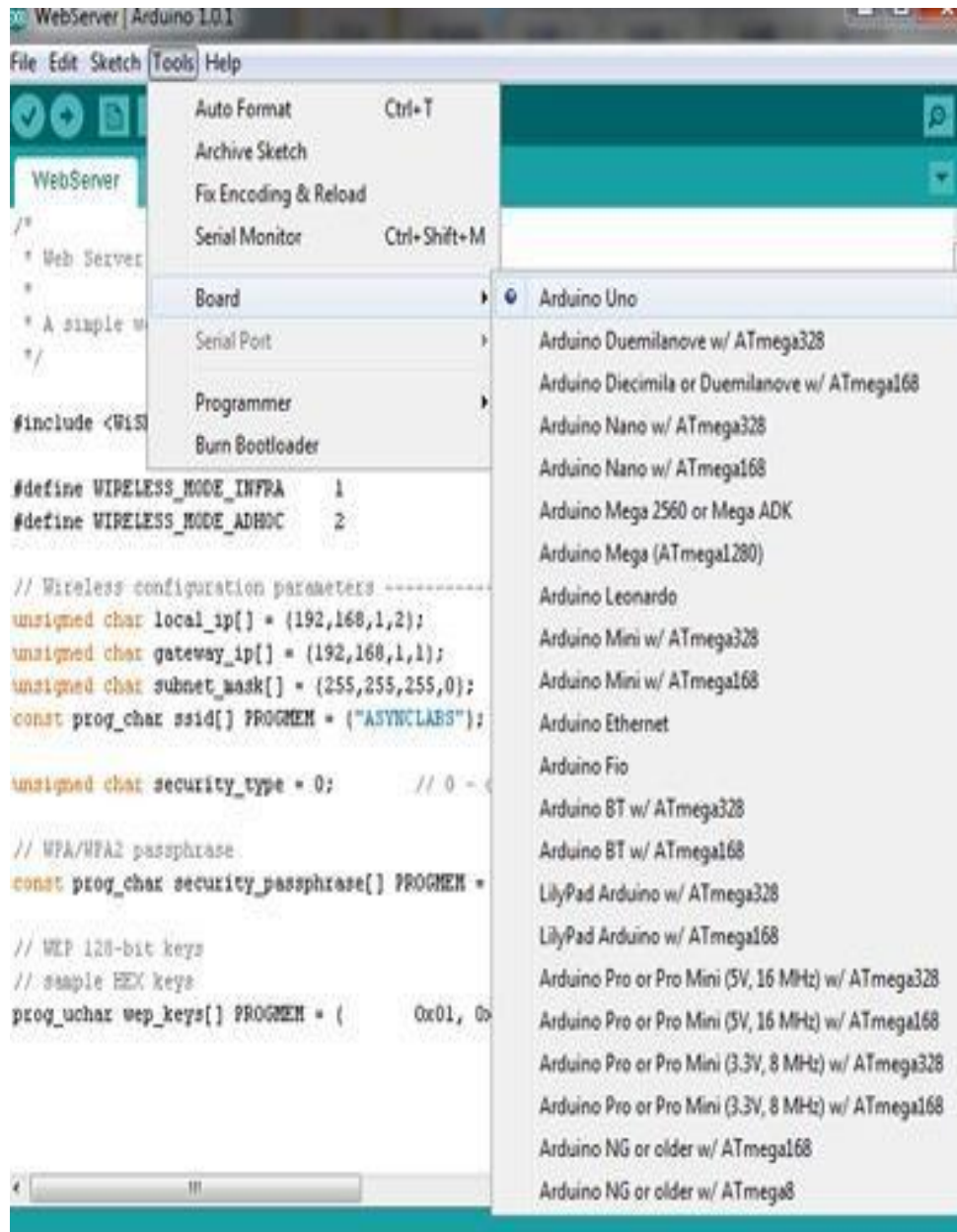
Launch Arduino by double click “Arduino” below



Example



Select the target board as “Arduino Uno”:



CHAPTER 8

METHODOLOGY WITH WORKING

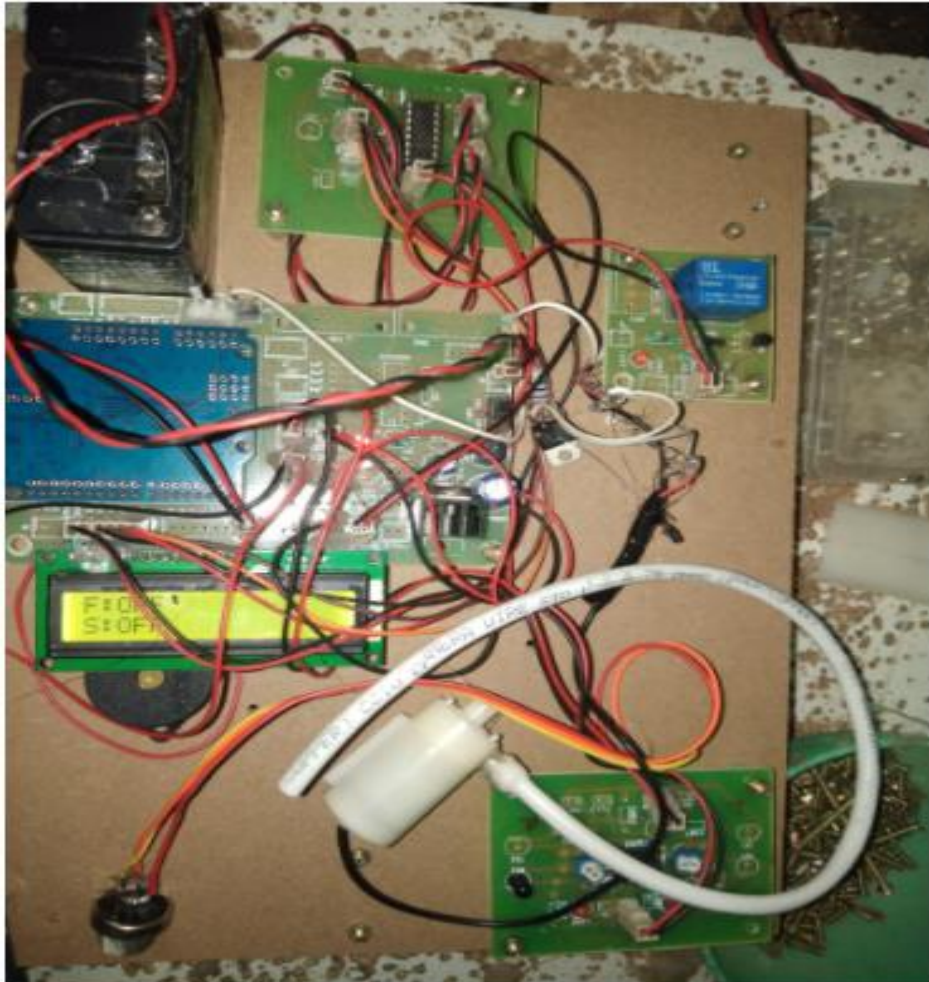


Fig 8.1 Hardware kit image

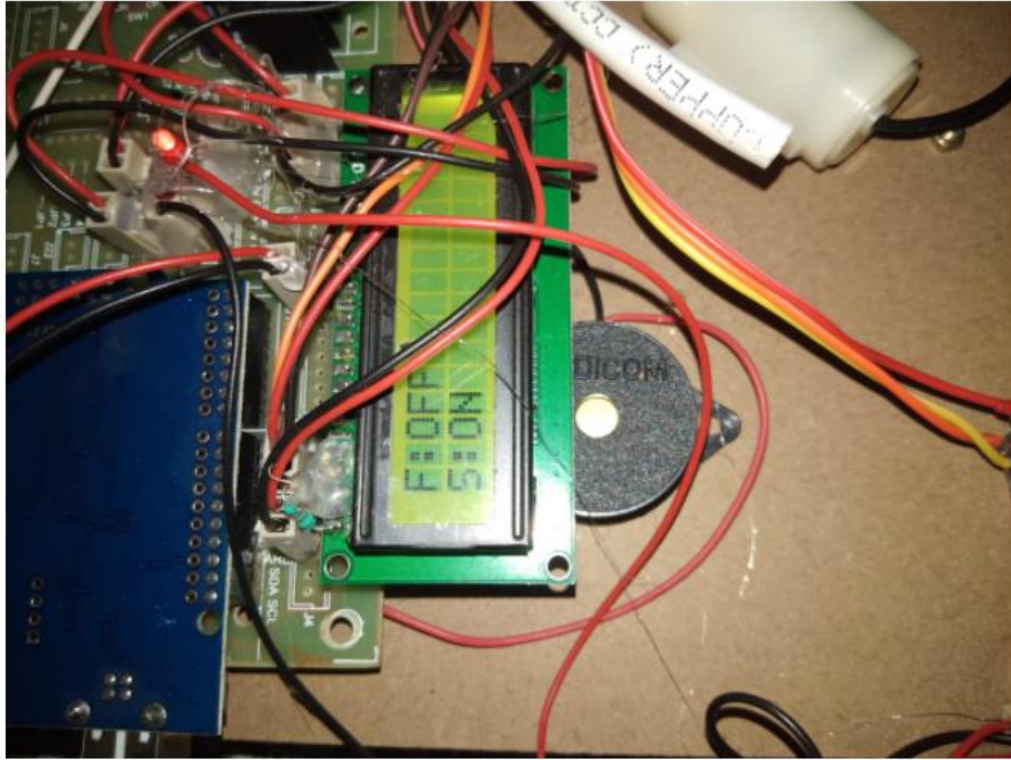


Fig 8.2 Smoke sensor detected time

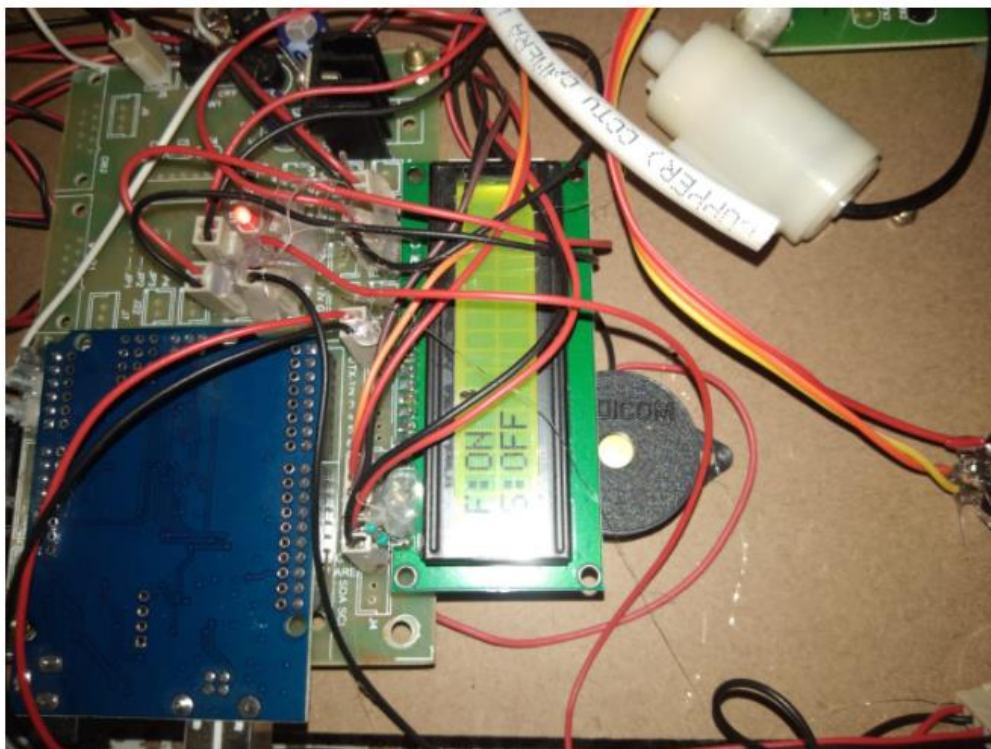


Fig 8.3 Fire sensor detected indication



A firefighting robot operates by autonomously or remotely navigating hazardous environments to combat fires and assist first responders. It utilizes a combination of sensors, actuators, and onboard systems for detection, navigation, and firefighting tasks. Sensors such as thermal cameras and gas detectors identify fire locations and hazards. The robot's mobility system, typically driven by DC motors, enables it to navigate terrain and obstacles.

Actuators control mechanical components like grippers or water cannons for firefighting actions. Centralized control systems process sensor data, make decisions, and execute tasks. Through this integrated approach, firefighting robots enhance safety and effectiveness in combating fires and supporting emergency response efforts.

CHAPTER 9

ADVANTAGES AND LIMITATIONS

ADVANTAGES

1. Easily track the missile.
2. To easily disposal of missile.
3. Buzzer indications also get here.
4. WSN also possible here.

APPLICATIONS

- Can be used in server rooms.
- Extinguishes fire where probability of explosion is high.
- Usable in Power plant control rooms, Captain bridges, Flight control centers.
- Disaster area monitoring and rescue.
- By inventing such a device, humans as well as property can be saved at higher rate with minimum damage caused by the fire.
- As instrumentation engineers, our task was to design and build a prototype system that could autonomously detect and extinguish a fire and also aims at minimizing the air pollution.
- The possibilities of fire are at any remote area or in an industry such as in garments go down, cotton mills, and fuel storage tanks, electric leakages may result in terrible fire & harm.
- The robot could be used to enter small spaces that are impossible to be accessed by a person.

CHAPTER 10

CONCLUSION

In many places such as schools, colleges our proposed project aims to develop an Arduino controlled fire fighter robot that can be used to extinguish fires through remote handling. By this we can avoid Fire accidents and also prevent manual intervention of fire extinguishing. We have tested it properly. The security system of the home and building contains firefighting robot security vehicle, F module. The main controller of the firefighting robot is a microcontroller.

We programmed the microcontroller to control the robotic vehicle to acquire flame sensor data, and run the vehicle towards fire by giving directions using commands given through the android mobile. Once the flame is detected by flame sensor, buzzer is activated. The water pump is activated in automatic mode. The robot is highly intelligent and likely able to be extended to a wider-range of applications and scenarios. Smoke detectors that set off fire alarms could be used to initiate the sound-activated robot. Pre-programming the robot with a floor plan and room locations would make it easy to find and extinguish a fire before humans are even aware of the situation.

Many of the methods used in this project, specifically in the navigation aspect, are still under research and development. Anyone interested in coming up with possible improvements could write their own algorithms given some programming experience in C++ and/or Python. It may be very challenging or even impossible to improve performance. T

here are smaller, lighter, microcontroller based robots, and are capable of very high speeds. But our robot is a versatile platform that could be used even just as a tool for navigation algorithm development.

FUTURE SCOPE

Some of interfacing applications which can be made are controlling home appliances, robotics movements, Speech Assisted technologies etc. A CO₂ booster can be attached to make it powerful extinguisher. In the present condition it can extinguish fire only in the way and not in all the rooms. It can be extended to a real fire extinguisher by replacing the fan by a carbon-di-oxide carrier and by making it to extinguish fires of all the room using microprogramming.

Also the robot could not be run through the batteries because at some conditions the current requirement for the circuit rises to about .8A which is very high and cannot be obtained using batteries. So we will try to develop the robot with many useful features like adding video camera, improvement in battery capacity, improvement in high weight etc...

There are several potential future enhancements that could be made to Arduino based fire fighting robots:

- Enhanced Sensing Technologies
- Intelligent Path Planning
- Wireless Communication
- Fire Analytics and Prediction
- Autonomous Recharging and Maintenance
- Collaboration with Emergency Services
- Integration with Building Systems
- Environmental Adaptability

CHAPTER 11

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